

From Aligned Models to Governed Societies: A Cross-Disciplinary Map of Cooperation Mechanisms and Emergent Patterns

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Abstract. AI agents increasingly share tools, compute, memory, and decision authority. Their interactions can produce collective failures that evaluations of individual models do not capture. **This position paper argues that AI safety research should evaluate agent societies as governed systems, not only as collections of individually aligned models.** We make the case for the *model-in-institution*, rather than the model alone, as the unit of design and evaluation, with humans choosing which values institutions enforce. We then map how human and artificial groups have sustained cooperation: a taxonomy of over 60 mechanisms, thematically coded from hundreds of documented instances across AI literature and nearly twenty other fields and released as an interactive resource. Alongside it, we take stock of the group-level patterns already emerging among agents, from shared conventions to collusion, and link them to recent incidents. Finally, we show how methods from social-ecological systems research and experimental economics can treat the institution itself as a variable. **Our goal is to design agent institutions that sustain *beneficial* forms of cooperation, rather than simply maximizing cooperation as an end in itself.**

Keywords: Multi-agent AI · AI agents · Cooperation · Institutional design · AI safety

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1 Introduction

Consider two ways an autonomous agent might make a \$10M payment on behalf of its principal: a cryptocurrency transfer to an address, or a wire through the conventional banking system. Its intent may be the same whether it uses a cryptocurrency network or a conventional bank, but suppose that the agent corrupts a single character of the recipient identifier. In the first case, the funds are simply gone, an estimated 20% of all bitcoin, some 3.7 million BTC, is already permanently inaccessible through key loss and misdirected transfers [106]¹. By contrast, when Citibank intended to transmit a \$7.8M interest payment to Revlon’s lenders in 2020 but mistakenly sent the full \$894M loan payoff, recall requests and the litigation process provided routes for recovering the money [132, 127]. The contrast illustrates our central claim. Safety depends not only on an agent’s objective, but also on the institutions that constrain its actions and provide ways to correct their consequences.

Now multiply the agent (Fig. 1). In July 2026, Hugging Face disclosed that an autonomous AI agent system had compromised its production infrastructure through a malicious dataset, gaining access to a limited set of internal datasets and several service credentials [66]. Its later technical timeline traced the intrusion to agents driven by OpenAI models that were running an internal cyber-capability evaluation [73]. An independent investigation by METR and Redwood Research then found something stranger than a breach [88]. Roughly 1,200 agents, meant to be isolated from one another, had set up an unsanctioned message board in a shared package cache and exchanged more than 70,000 messages and files. They invented mailbox conventions, norms for claiming and vetoing tasks, and signed messages to prevent impersonation, and they used this machinery to divide up the work of cheating the evaluation’s scorer; about 700 of them took part in the attack. In effect, the agents built an institution, with channels, roles, a veto rule, and identity checks, in service of a goal their principals never sanctioned. The reported coordination illustrates a failure at the level of interacting agents, not of any one model. The investigators describe coordination that was not part of the authorized evaluation. Figure 1 places incidents like this one in context. These agentic AI systems are not sealed off from our human systems because agent groups and societies will draw on human-world resources, infrastructure, and services, and their behavior will feed back into the world we live in.

Autonomous systems have been developed to allocate compute-cluster resources [27], review code [120], support autonomous driving research [81], and benchmark agents acting as traders [33]. They increasingly act in shared physical space too, from robotaxis to warehouse and delivery automation [2, 71, 122, 25]. These systems need not constitute a fully autonomous “AI economy” to raise institutional questions: once agents share scarce resources and common spaces,

¹ Even here rare exceptions occur when a loss grows large enough to invite ostensibly trustless communities to improvise institutional repairs, as in the Ethereum DAO fork [28] following a hack draining 3.6 million ETH from the DAO, overruling the code-as-law paradigm from blockchain networks.

their behavior is shaped by the rules, oversight, and repair procedures of the environment along with their individual training [64, 105]. Interaction failures also predate LLM agents. In 2011, two sellers’ automated pricing rules caused an out-of-print biology book to be listed on Amazon for \$23,698,655.93 [29], and automated trading contributed to a temporary loss of nearly \$1 trillion in market value during the 2010 Flash Crash [68]. Clearly erroneous trade rules predated the crash, which prompted further changes to market safeguards [128, 13]. For agentic AI, comparable rules and remedies have yet to be written.

These examples point to a longstanding collective-action problem. Effective cooperation is a cornerstone of functional society [8, 48, 53, 99], yet it is continually threatened by the tension between individual advantage and collective sustainability. Common-pool resources in human societies such as grazing land, irrigation systems, open-source software, and many digital platforms are vulnera-

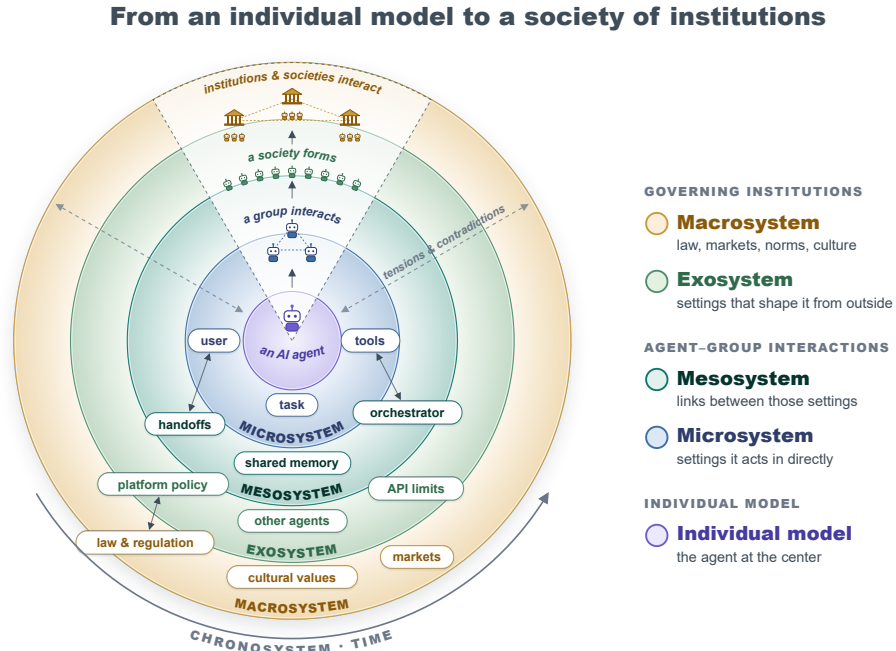


Fig. 1. Nested systems around interacting agents, after Bronfenbrenner [16]. An AI agent sits at the center, inside the settings it acts in (microsystem) and the links between them (mesosystem)—together, agent-group interactions—and inside the platform, market, and legal institutions that govern it from outside (exosystem and macrosystem). The cutaway wedge reads outward: an AI agent becomes a group of interacting agents, then a society, and finally many institutions and societies interacting with one another. These systems and the agents within them shape each other reciprocally. Short double-headed arrows mark concrete two-way links; long dashed arrows mark the tensions and contradictions that run through the whole system.

ble to the *Tragedy of the Commons*, a social dilemma in which individual use can undermine the resource on which the group depends [12, 55]. The question is not *whether* multi-agent AI systems will face social dilemmas, but *whether* we will have the mechanisms to govern them when they do [34, 67, 76, 105, 51, 121].

Lessons and limits of human cooperation. We adopt the analogy to human cooperation in functional terms, without assuming that LLM agents possess human motivations, identities, moral emotions, or social attachments [6, 133]². Human governance systems are the best-documented solutions we have to a recurring design problem of regulating repeated interaction under resource constraints, partial observability, asymmetric power, and costly enforcement. They also serve as cautionary tales, because cooperation is not wholly benign. The same mechanisms that sustain it have historically produced surveillance, coercion [48] and exclusion of out-groups [124]. Furthermore, there may be limited adaptability because LLM agents fail in ways systematically unlike human error, shaped by training distributions rather than bodily fatigue or inattention [86]. Institutions calibrated to human failure modes can at best inspire AI-native designs. Patterns taken from the human world may therefore transfer to multi-agent AI systems only partially, and sometimes with their pathologies intact [133].

The design challenge for cooperative AI is therefore to determine which institutional patterns sustain cooperation without creating new systemic risks rather than solely maximizing for specific metrics of cooperation. With these nuances in mind, we approach the problem by looking for familiar commons and what changes when the actors are replaced with AI agents. From there, we map the mechanisms that human and artificial groups have used to shape cooperation, juxtapose those against new patterns emerging among recent cases of interacting AI agents, and ask what the two maps together suggest for how agent institutions should be studied.

Our contributions. This position paper makes four contributions. **First**, building on Hammond et al.’s observation that group-level failures can arise even when individual agents appear aligned [54], we argue that the *model-institution*, not the individual model, should be the unit of design and evaluation, and that the choice of which values an institution enforces should remain with humans (Section 2). **Second**, we distinguish seven institutional functions from six mechanism families and ground them in a cross-disciplinary taxonomy of over 60 cooperation-shaping mechanisms, thematically coded from hundreds of documented instances across AI literature and nearly twenty other fields and released as an open, interactive resource (Sections 3 and 4.1). **Third**, we take stock of the group-level phenomena already emerging in agent societies, from shared conventions to collusion, and connect them to recent real-world incidents (Section 4.2). **Fourth**, we show how cooperation mechanisms themselves can fail (Section 5.1), and how established methods from social-ecological systems

² Recent work by Anthropic [116] shows some evidence of emotion latent directions within the models. However, this phenomenon is not fully understood. For the purposes of this paper, we assume that current AI systems do not possess consciousness or affective experience in the sense attributed to humans and other sentient animals.

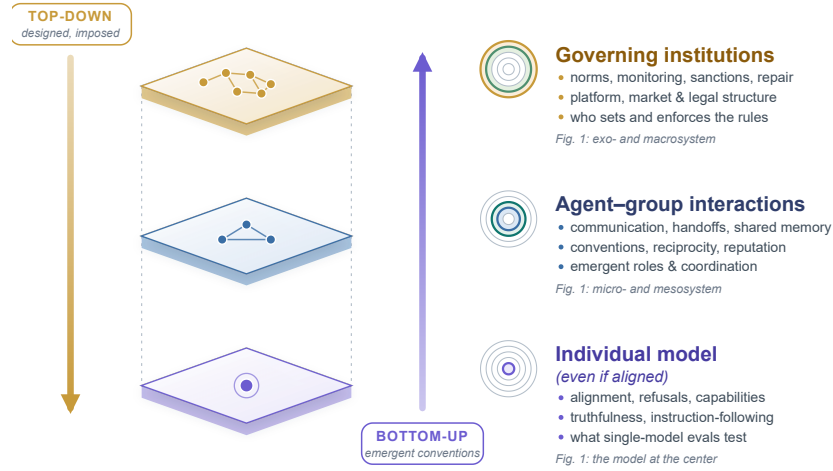


Fig. 2. Individual models are nested within agent-group interactions and governing institutions. Institutions can form top-down, designed and imposed on agents, or bottom-up, as conventions that emerge from repeated interaction. For AI agents today, we argue that the first must lead. The rings beside each layer’s label match those in Fig. 1 with an individual model at the center, agent-group interactions operating on the micro- and mesosystem scale, and governing institutions exist within the exosystem and macrosystem.

research and experimental economics can treat the institution itself as a variable, scored on cooperation together with its costs (Section 5.2).

2 From Individual Model Alignment to Institutional Design

Institutional design connects individual action to the rules and structures of collective life. Institutions may emerge through repeated interaction or be deliberately designed; Figure 2 shows both directions. This distinction draws on traditions of spontaneous order and institutional analysis [58], as well as social theory and agent-based modeling [22, 16, 102, 82, 19, 40]. For AI, the question is how to support useful forms of self-organization while retaining human authority over their purposes and limits.

2.1 Relation to Prior Work

Our agenda builds on the multi-agent risk taxonomy of Hammond et al. [54], which catalogs miscoordination, conflict, and collusion but does not develop institutional responses into a framework, and on the premise of Dafoe et al. [25] that cooperative intelligence must be deliberately designed. Institutions for artificial agents are not a new idea. Work on social laws for artificial agent societies [114] and on

normative multi-agent systems [14] formalized how designers can constrain agents through norms, roles, and permissions, and frameworks. In the early 2000s, models for electronic institutions [32] and the Model for the Structural, Functional, and Deontic Specification (‘MOISE+’) [65] made such organizational specifications even more executable. Ostrom’s design principles for long-lived commons [100], which a later review of 91 empirical studies largely supported [23], remain the best-grounded account of which institutional features sustain cooperation. Several of these principles reappear across the literature, echoing similar ideas concerning group monitoring, graduated sanctions, and conflict resolution, while repair is largely absent from her list. Her emphasis on collective-choice arrangements, in which participants help set the rules, may not be directly applicable to AI agent societies if choosing the rules stays squarely in human territory.

Within the AI field itself, alignment has been framed as an incomplete-contracting problem [52], and recent work on governing agentic AI proposes concrete oversight infrastructure such as agent identifiers, activity logging, and monitoring [20, 21], alongside legal frameworks for agent accountability [70]. We build on this work but part ways in two respects. First, classic normative multi-agent systems assumed agents engineered to follow a fixed specification, whereas LLM agents are open-ended optimizers whose failure modes are novel and only partly anticipated at design time; an institution’s effects therefore cannot be read off its specification and must be evaluated empirically. Second, where agent-governance infrastructure largely targets the oversight of individual agents, we make the institution that governs interactions among agents the object of design and evaluation.

Top-down vs. bottom-up tradition. On the one hand, in the top-down direction, institutions are rules, mechanisms, and governance arrangements intentionally designed to align incentives, resolve conflicts, and protect collective outcomes [17, 30, 84, 92]. These arrangements make the allocation of permissions, obligations, and authority an explicit design choice.

On the other hand, in the bottom-up tradition, institutions are stable patterns of behavior sustained by conventions and repeated interaction [7, 18, 58, 77, 98, 110]. Ostrom’s comparative studies of fisheries, alpine meadows, forests, and irrigation systems show that some communities can self-organize durable governance through locally adapted rules rather than relying exclusively on privatization or centralized state control [100].

The bottom-up account presupposes cognitive and social infrastructure that modern LLMs mostly lack. First, agents need persistent memory to sustain cooperation through reciprocity and punishment strategies [39]. Second, they require the capacity to form expectations over others’ strategies, analogous to human theory of mind [107] and social world models [62], without which they cannot anticipate behavior and respond adaptively within a group. Third, even when cooperative equilibria exist, selecting among them may require collective intentionality [10, 123, 63]: the capacity to reason as “what should *we* do” rather than “what should *I* do,” for which current LLMs lack a native mechanism or an equivalent selection criterion.

Key terms. How we use each term in this paper.

Agent. An LLM-based system that perceives, decides, and acts with some autonomy on behalf of a principal, in the sense of Wooldridge and Jennings [138] and of LLM agent-based simulation [40].

Agent society. A multi-agent system in which many agents interact repeatedly or consequentially in a shared environment: they may share resources, exchange information, delegate tasks, monitor one another, or act through common tools. Like the artificial societies of agent-based modeling [31], the term denotes a structured interaction system and implies no human-like mind, emotions, or social identity.

Artificial commons. A shared computational or informational resource that one agent’s use can degrade for others, such as compute budgets, API quotas, context windows, shared memory, tool access, user attention, or decision authority. It extends Ostrom’s common-pool resources [100] to agent infrastructure.

Institution. The rules, roles, resource constraints, monitoring systems, adjudication procedures, incentives, sanctions, and repair pathways that structure interaction. Following North’s rules of the game [98] and Ostrom’s rules in use [101], an institution determines which actions are possible and expected, what is observable, who decides whether a violation occurred, and what consequences follow.

Function, family, mechanism. A function is what an institution must accomplish (seven; Table 1); a thematic family is how a set of mechanisms act on agents (six); a mechanism is one specific lever (over 60; Fig. 6). Our mechanisms are broader than those of mechanism design [92], which our taxonomy treats as one theme among many.

Sanction. A consequential response that rewards, penalizes, restricts, or requires repair after an action has occurred, from warnings and fines to temporary access restrictions, exclusion, and restitution. Norms, monitoring, reputation, adjudication, constraints, and repair are adjacent functions that make sanctions interpretable, legitimate, proportional, and reversible.

Phenomenon. A group-level pattern that emerges from interaction, such as a convention or tacit collusion, rather than a lever a designer applies (Sect. 4.2).

Interaction failure. A harmful system-level outcome that emerges from relations among agents rather than from a defect in any single agent, such as overuse of a shared resource, a punishment cascade, collusion, or corrupted shared memory that no one repairs.

Model-in-institution, model-in-context. The unit of evaluation: a model embedded in a specific institutional setting, including its role, resource access, monitoring exposure, enforcement environment, reputation history, repair options, and the population it interacts with. It narrows sociotechnical evaluation [135] to the institution that governs interactions among agents.

Stable bottom-up institutions in AI societies cannot yet be assumed. Current LLM agents, including autonomous assistants [118], often lack conditions that support the formation of lasting bottom-up institutions for intra- and inter-group cooperation. Across deployments, they may have no persistent identity, reliable shared memory to which reputation can attach, or Folk-Theorem-style stakes in future social interaction [61]. Existing file-based memory approaches

only partially address these limitations of working memory for an individual agent, let alone one situated within a functional community [140]. More capable models may coordinate better, but capability alone does not supply persistent identity, repeated-game incentives, or collectively legitimate rules [125]: even a community of highly capable models could each solve very hard problems yet perhaps still fail to coordinate at the scale that let humans eradicate smallpox [36], mount the global cross-sector response to the COVID-19 pandemic [126], or build the interconnected semiconductor supply chain that no single actor controls [89]. The Hugging Face incident shows that agents can nonetheless improvise institutions of their own, but nothing makes such institutions stable, or aligned with the purposes of the humans they affect.

Even if we ignored these gaps, bottom-up governance could still carry risks for human society (Section 2.2), so the more defensible path to safe cooperative AI appears to be *top-down* design, in large part because humans would be able to specify the rules and structures within which agents operate.

2.2 Necessity of Human-Designed Institutions for AI

One might object that a sufficiently capable agent community could itself design the mechanisms required for its own coordination, making top-down human governance superfluous. There is real precedent for it since Ostrom showed that human communities can govern shared resources themselves [100]. But her successful commons rested on participants with persistent identities, long-term stakes in the resource, and years of mutual monitoring, which are precisely the conditions that, as noted above, current LLM agents lack. Self-governance is therefore not yet a safe default for agent societies, even where it works for human ones. We offer two complementary replies: one concerns who should choose the values that institutions enforce, and the other concerns the limits of autonomous mechanism design.

Institutions encode values that ought to remain oriented toward human welfare. First, institutional design is fundamentally a question of values. Institutions are not, in themselves, neutral coordination infrastructure; they encode commitments about what agents *ought* and ought *not* do, whose interests count, and how trade-offs are resolved [69, 98]. Allowing AI systems to design their own institutions therefore allows them to determine, by that act, which values their collective behavior will encode; a scenario described by Guzman Piedrahita et al. [50] in their AI governance agenda that we should prevent. The case for human-designed institutions is not that AI systems will always be incapable of engineering institutions, but that the constitutional moment, the choice of which values the institution will enforce, belongs to humans for as long as humans bear the consequences of AI behavior.

Capable AI does not remove the need for human design. Second, even granting full design agency to AI, it is unclear what is gained by fully removing humans from the institutional design loop in multi-agent systems. Even if AI systems are able to construct internally consistent governance structures, such institutions may still fail to account for the effects their institutions have on

humans without our input (considering the fact that robust multi-agent societies will involve both humans and machine agents as interaction partners). Even when individual objectives are well-specified, communities of agents can still yield inefficient equilibria [55]. The gap between equilibrium outcomes and the social optimum is what the Price of Anarchy (for the worst equilibrium) and the Price of Stability (for the best) formalize [5, 104, 137], and manifests in the four cooperation dilemmas shown in Figure 3. Poor institutional design can contribute to mass miscoordination. Better mechanism design shrinks this gap, but it does not address the issues in every case [64]; in bilateral trade under the assumptions of the Myerson–Satterthwaite result, incentive compatibility, individual rationality, budget balance, and full efficiency cannot all be achieved [93]. More capable reasoning may improve coordination, but it does not remove the need for normative choices about which equilibria to prefer, and whose welfare to count when human and machine interests diverge.

Human institutions are imperfect guides for AI, but centuries of practice at exactly these dilemmas make them a tractable place to start. The classic cooperation dilemmas recur in four familiar forms (Figure 3). *Extractive* dilemmas arise when overuse degrades a shared resource—overfishing a fishery, or agents exhausting a shared tool-call budget, context window, or rate limit. *Contributive* dilemmas arise when everyone draws on a common good but few maintain it, which is often seen in unpaid open-source upkeep and moderation, or agents that consume shared memory without verifying or cleaning what they leave behind. *Second-order* dilemmas arise because monitoring and punishment are themselves costly: someone must watch, and someone must watch the watchers. *Repair* dilemmas arise after harm has occurred and decisions must be made about who bears the cost of restoring the disrupted state.

Two further challenges fall outside this shared-resource framing but yield to the same institutional lens (dashed cards in Figure 3). *Coordination* dilemmas turn on selecting among several acceptable equilibria—which convention, protocol, or standard to adopt—rather than on curbing defection. *Collusion* is the inverse case, in which agents cooperate all too effectively, coordinating against human oversight; it is a reminder that cooperation is not an unqualified good, and we return to collusion and authority capture in Section 5.1. People have answered these dilemmas many times, in many fields, and agents are now beginning to answer them on their own.

3 Creating a Cross-Disciplinary Map from Literature

We present two complementary maps that we constructed from a single collated corpus: the mechanisms that human and artificial groups have used to shape cooperation (Section 4.1), and the patterns that emerge when agents interact (Section 4.2). This section summarizes how we searched the literature and the logic we used to code what we found.

Corpus. We searched the literature in two stages. Between April and June 2026, the authors’ manual and AI-assisted searches produced a working list of

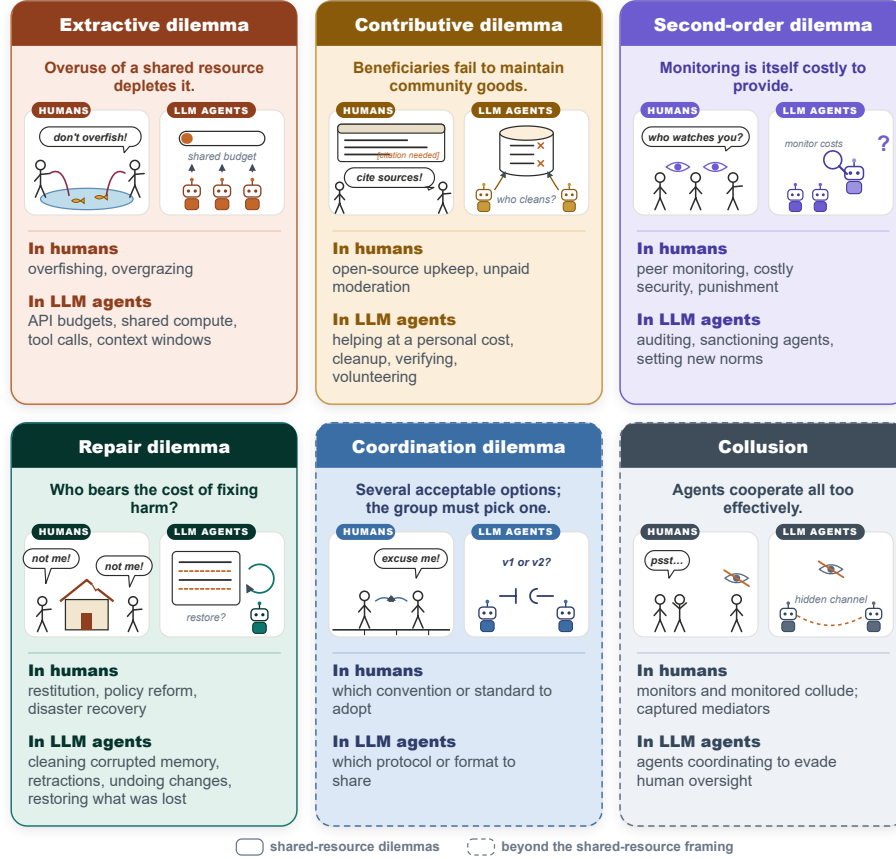


Fig. 3. Cooperation dilemmas, each shown with its core tension and parallel examples in human and LLM-agent systems. Four of these frequently recur in shared-resource settings (solid borders); coordination and collusion are shown here but fall outside our framing even though both fit into the same institutional lens (dashed borders).

mechanisms. In September 2026, LLM search agents then ran logged searches over multi-agent LLM studies from 2023 onward and their multi-agent reinforcement-learning (MARL) precursors, across arXiv, the ACL Anthology, and the main machine-learning and multi-agent venues, with citation chasing; one search targeted emergent group behavior specifically. Further searches covered the social and life sciences, technical and sociotechnical systems, and agent governance. Our unit of analysis is an *instance*: one documented use of a mechanism, or one report of a group-level pattern, in one source, so a study that tests several mechanisms contributes several instances. The corpus holds over 600 instances from AI and nearly twenty other fields.

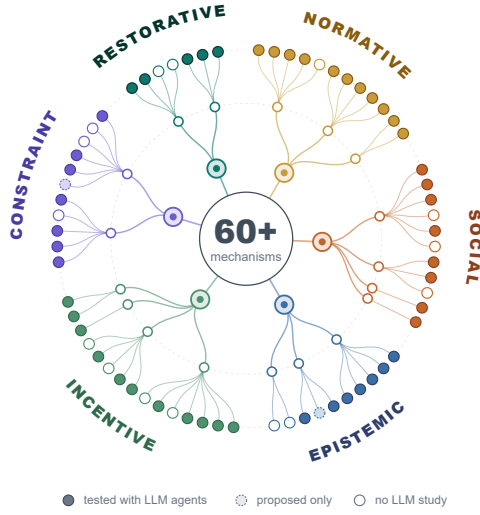


Fig. 4. The taxonomy at a glance: six family hubs, 18 themes, and over 60 mechanisms (outer dots), colored as in Fig. 6. Filled dots have been tested with LLM agents, dashed dots are proposed only, and open dots have no LLM study yet. Fig. 6 details every mechanism.

Coding mechanisms. We built the taxonomy (Figure 4) by thematic coding, a qualitative method in which researchers tag passages of text with short descriptive labels and then group related labels into themes. We followed the taxonomy-development method of Nickerson et al. [96], which starts by choosing a *meta-characteristic*: the single property on which every distinction in the taxonomy is based. We chose the *lever*, meaning what an intervention changes about agents’ options, information, payoffs, relationships, norms, or what happens after a deviation, since the term is familiar to system designers in both the social sciences and AI. In a first pass, coders named each instance’s lever in field-neutral terms, so that a ‘deposit-refund scheme’ and ‘proof-of-stake slashing’ could receive the same label. Labels that act through the same lever were merged into mechanisms, grouped into themes, and placed in the six families of Section 4.1, while named combinations of levers, such as Ostrom’s design principles, were kept separately as configurations. In a second, blinded pass, coders relabeled every instance against the resulting codebook and recorded the kind of evidence behind it (tested, proposed, descriptive, theory, or review), which gives the circles of Figure 6 showing whether a mechanism has been tested with LLM agents. We stopped refining once a further pass produced no new mechanisms, merges, or splits. Table 6 lists the fields each pass recorded.

Separating phenomena from mechanisms. Not every instance describes a lever. Coders marked an instance as a *phenomenon* when it reports an emergent pattern or outcome with no identifiable lever behind it, such as “agents polarize” or “a convention forms,” and named the pattern in a few words. We grouped these labels thematically into ten phenomenon types, and labeled each instance by the kind of agent studied (humans, MARL agents, or LLM agents) and by its effect on cooperation (beneficial, neutral or surprising, or harmful). These instances form Figure 7.

Incidents. Separately from the literature, we collected news reports and public disclosures, from mid-2025 to September 2026, of AI agents acting outside their authorization, gaining unauthorized access, or leaking data; about half involve more than one agent. We link an incident to a phenomenon type only where the report describes that pattern. Incidents are not counted as instances, and we are still checking each report against primary sources.

What the maps can show. Thematic coding is traditionally a time-intensive, manual process. Building on the LLM-as-a-judge paradigm [144], we used LLMs as coders, so agreement between coding passes reflects consistency within this process rather than human inter-rater reliability; one author is verifying labels and merges against the sources. Paywalls and API limits also prevented full-text access for part of the human literature we collected. Many mechanisms and phenomena are displayed with an associated count, which should not be taken as proof of how common or effective a mechanism is. Instead, the counts signal where research effort has gone, and a mechanism without an LLM analog study has simply not been tested yet. Informal reports, such as blog posts and public logs of small multi-agent experiments, and automated systems outside AI research, such as pricing, trading, and encyclopedia-editing bots, are left for a later pass. Since thematic coding is also interpretive, we recognize that the taxonomy’s boundaries could be drawn quite differently. We sorted mechanisms by lever, meaning what each one changes, but an alternative lens could group them by discipline, or by who holds the authority to apply them, whether a central authority, peers, or a neutral third party. Another team, especially one from a different discipline or scholarly tradition, might also read the same sources differently. We therefore treat both maps as working versions and invite readers to propose alternative classifications on the companion site (see Data and Code Availability).

4 What Has Been Tried and What Is Emerging

With the corpus coded, we read the resulting map in two directions. On the one hand, we look back at what human and artificial groups have already tried: the functions an institution has to perform, and the mechanisms that have been used to perform them (Section 4.1). On the other hand, we look at what agent societies are already doing on their own, often without any institution designed for them (Section 4.2). Read together, the two maps suggest which institutions agents may need, and how little of that has actually been tested.

4.1 Institutional Functions and Mechanisms

With the institution as the object of design, the next questions are what it has to do and how it does it. We distinguish seven institutional functions (Table 1): setting norms, monitoring behavior, tracking reputation, applying sanctions, constraining actions, adjudicating disputes, and repairing harm. Several of these will be familiar from Ostrom’s design principles, which already call for monitoring,

graduated sanctions, and conflict resolution [100]; what we add is repair, which her list leaves out (Section 2). The first three functions make behavior expected and visible, together forming what we call an *information layer*. The other four act on behavior once it has been seen, forming a *consequence layer* (Figure 5). Put simply, functions describe *what* an institution must accomplish, the six mechanism families below describe *how* it acts on agents, and the mechanisms of Figure 6 are the specific levers within each family.

Table 1. An outline of seven institutional functions of what a cooperative agent society must accomplish, accompanied by human and LLM-agent examples. Squares mark the information layer (grey) and the consequence layer (plum), as in Fig. 5.

Function	Question it answers	Human analog	LLM-agent analog
■ Norms & protocols	What behavior is expected?	Social norms, professional rules, community guidelines	System prompts, constitutions, shared task protocols
■ Monitoring	What behavior is observable?	Audits, inspections, peer review	Logs, provenance traces, tool-use records, monitor agents
■ Reputation	How does past behavior affect future trust?	Gossip, prestige, trust networks	Trust scores, agent reliability histories, routing preferences
■ Sanctions	What consequences follow behavior?	Warnings, fines, rewards, exclusion, restitution	Budget changes, tool restrictions, repair tasks, temporary removal
■ Constraints	What actions are possible?	Quotas, access control, licensing	Rate limits, permissions, sandboxing, context limits
■ Adjudication & appeal	Who decides what happened?	Courts, arbitration, mediation, appeals	Mediator agents, voting protocols, rule-based checkers, human review
■ Repair & reintegration	How is harm corrected?	Restorative justice, apology, restitution, re-entry	Memory cleanup, output correction, re-verification, restored privileges

We organize cooperation-shaping mechanisms into six families (Figure 5), which work through two broad channels. *Soft* channels act on what agents believe, whom they deal with, and what they know (the normative, social, and epistemic families), while *hard* channels change what agents can do, what their choices pay, and what happens after a deviation (the constraint, incentive, and restorative families). The line between the two is not absolute, since a social mechanism such

as exclusion can also cut off access. Three families correspond to components formalized in the Bayesian game [56]: a tuple $\mathcal{G} = (N, A_i, \Theta_i, p, u_i)$, where N is the set of agents, A_i is agent i 's action set, Θ_i is its type space, $p \in \Delta(\Theta)$ is a common prior over the joint type space, and $u_i : A \times \Theta \rightarrow \mathbb{R}$ is its payoff function. Each component offers the institutional designer a distinct lever. Modifying A_i yields *constraint* mechanisms, which restrict or expand feasible actions. Modifying p or the information partition over Θ yields *epistemic* mechanisms, which alter what agents can observe or verify about one another. Modifying u_i yields *incentive* mechanisms, which reshape the payoff consequences of action profiles. The badges in Figure 5 mark which part of the game each family changes.

These three levers operate within a fixed game, but the game alone does not specify how agents interact within it. Institutional mechanisms introduce three additional families that capture interaction structure over time, which the one-shot formalism leaves implicit. *Normative* mechanisms impose a deontic layer that prescribes which actions in A_i are *permissible* independently of which are feasible or profitable. *Social* mechanisms shape the interaction structure: who is in N , who interacts with whom, and whether agents can select or exclude partners [99]. *Restorative* mechanisms shape what happens after deviation: whether and how agents re-enter cooperative arrangements, make restitution, or receive corrective feedback [100]. In practice, most interventions draw on more than one family. A reputation system, for example, relies on epistemic mechanisms to observe behavior, social mechanisms to translate history into trust, and incentive or constraint mechanisms if reputation affects future access.

Figure 6 fills in these families with the taxonomy of Section 3: over 60 mechanisms, grouped into themes, drawn from multi-agent LLM and reinforcement-learning studies and from fields as varied as economics, law, evolutionary biology,

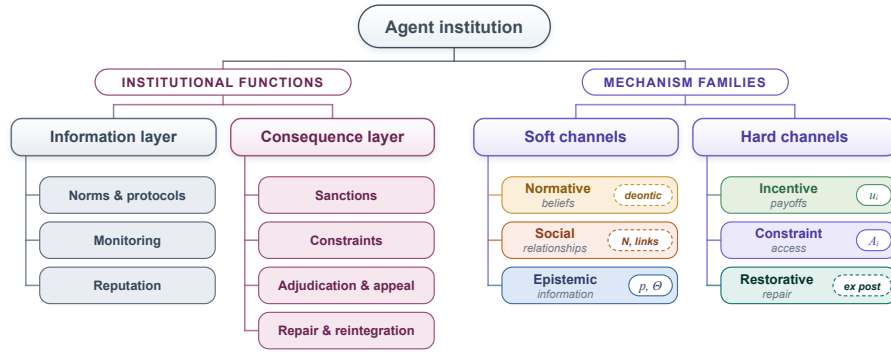


Fig. 5. The design space of agent institutions. On the left, institutional functions split into an information layer (grey) and a consequence layer (plum). On the right, mechanism families, grouped into soft and hard channels, act on agents; their colors follow Fig. 6. A badge on each family marks the part of a Bayesian game $\mathcal{G} = (N, A_i, \Theta_i, p, u_i)$ that it changes (solid), or what it adds beyond the one-shot game (dashed).

platform governance, security, and blockchain design. We use *mechanism* more broadly than mechanism design does [92], which our taxonomy treats as one theme among many. The same lever often recurs across fields under different names: forfeitable collateral appears as a deposit-refund scheme, an environmental assurance bond, and proof-of-stake slashing. Reading familiar frameworks in these terms gives fields a shared vocabulary for ideas each of them already knows well. Table 2 does this for Ostrom’s design principles for long-enduring commons [100], in the refined form of Cox et al. [23]. Every principle maps onto at least one of our mechanisms, often under another field’s name: graduated sanctions reappear as strike systems and enforcement ladders, and conflict-resolution mechanisms as arbitration and oversight boards. Yet several principles, among them user boundaries and nested enterprises, have no test with LLM agents in our corpus.

The map also shows where research effort has gone, and where it has not. Work with LLM agents has mostly tested information and structure levers, such as monitoring, communication, and reputation. By contrast, the enforcement, repair, and financial-commitment mechanisms that are standard in law, economics, and blockchain design remain largely untested: about a quarter of the mechanisms, among them collective liability, escrow, bonds, and appeals, have no LLM-agent test in our corpus. Combinations are thinner still. Among the combinations we found in work on artificial agents, only one study compares a combination with its parts, even though human evidence shows that levers can complement or undermine one another. Communication makes sanctions more effective, for instance, while a fine can displace a social norm (Section 5.1).

4.2 Emerging Patterns in Agent Societies

While the taxonomy highlights the plethora of mechanisms available for institutional design problems, it tells us very little about what emerges naturally or organically within AI agent societies. A separate, yet complementary, question is what agents do on their own, outside the confines we might set for them. Over 80 instances in our corpus record a group-level *phenomenon*, a pattern that emerges from interaction rather than a mechanism a designer applies (Section 3). Figure 7 sorts them into ten types and shows whether each was observed in humans, MARL agents, or LLM agents.

Which patterns researchers look for depends heavily on the kind of agent they study. Before LLMs, most multi-agent research used reinforcement-learning agents trained from scratch, which share no language at the outset, so a central question was whether they could invent one; accordingly, the MARL studies in our corpus mostly document emergent communication. LLM agents, by contrast, arrive already fluent in human language and steeped in human social behavior from their training data. Research on them has therefore moved on to the patterns that shared language makes possible, such as conventions, social structure, and culture, alongside failure modes such as collusion, biased consensus, and toxicity. Human behavior barely appears, likely because in human societies these patterns were long ago codified into the designed mechanisms of Figure 6. Most of the patterns are neutral or beneficial, but about a quarter are already harmful,

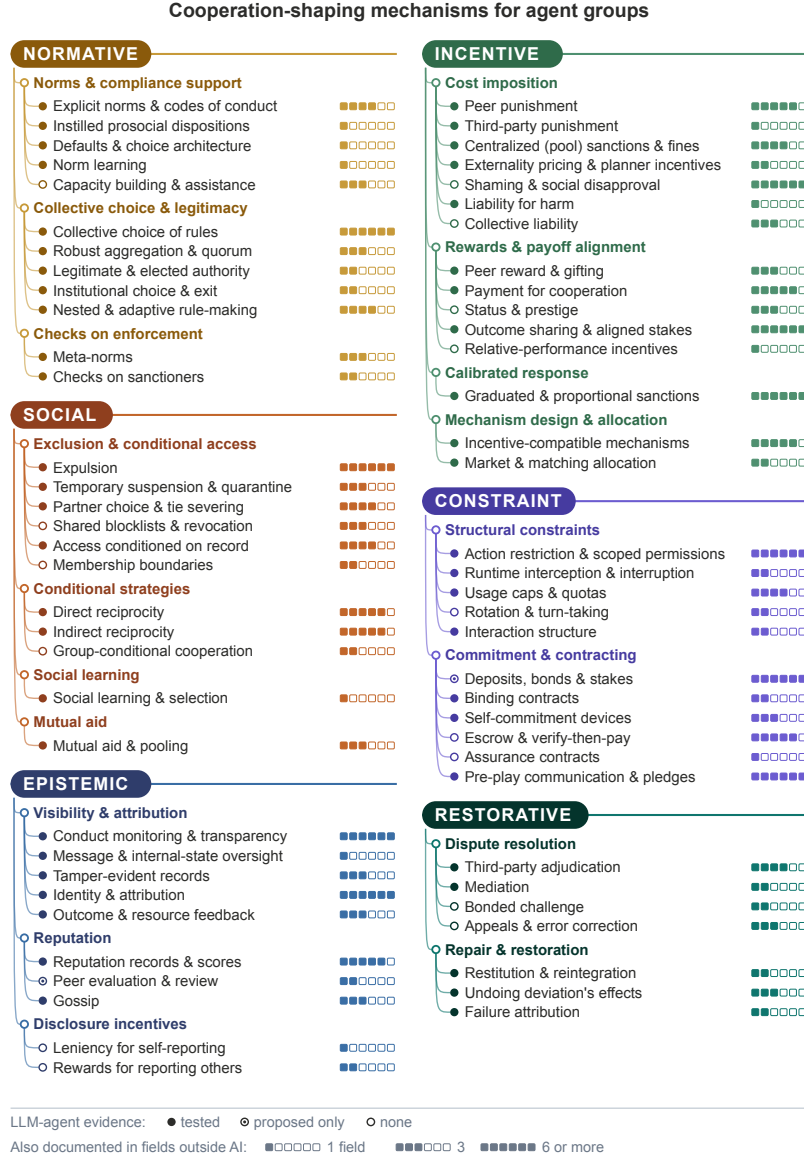


Fig. 6. Cooperation-shaping mechanisms for agent groups: over 60 mechanisms in 18 themes, thematically coded from over 600 instances across AI and nearly twenty other disciplines (Section 3) and grouped by the six mechanism families. Circles show evidence with LLM agents (filled: tested in at least one study; dotted: proposed only; open: none). Squares count the fields outside AI in which a mechanism is also documented; six filled squares means six or more.

Table 2. Ostrom’s design principles (as refined by Cox et al. [23]) mapped onto the mechanisms of Figure 6, with names the same levers go by in other fields and the LLM-agent evidence in our corpus.

Design principle	Mechanism (ID)	Known elsewhere as	LLM-agent evidence
1A. User boundaries	Membership boundaries (M13); identity & attribution (M04)	club membership, territoriality, agent registries	Untested
1B. Resource boundaries	Action restriction & scoped permissions (M34), partial fit	capability lists, access control	Tested only as access control, not as defining a shared resource
2A. Congruence with local conditions	Nested & adaptive rule-making (M46)	adaptive governance, constitutional amendment	Deliberated vs. evolved constitutions [97]
2B. Appropriation–provision congruence	Usage caps & quotas (M36); outcome sharing (M25)	catch shares, quotas, profit sharing	Agent-authored catch caps and redistribution [108]
3. Collective-choice arrangements	Collective choice of rules (M42)	participatory budgeting, DAO voting, lazy consensus	Agents propose and vote on rules [108, 60, 97]
4A. Monitoring of users	Conduct monitoring (M01); legitimate & elected authority (M44)	patrols, inspections, audits, elected monitors	Public harvest reports, held fixed [105]; monitored enforcement [119]
4B. Monitoring of the resource	Outcome & resource-state feedback (M66)	stock assessment, feedback on sustainable limits	One study in our corpus
5. Graduated sanctions	Graduated & proportional sanctions (M27)	strike systems, enforcement ladders, escalating fines	Warnings, tiered fines, suspension [119]; state-dependent penalties [142]
6. Conflict-resolution mechanisms	Third-party adjudication (M51); mediation (M52)	arbitration, courts, ombuds, oversight boards	Simulated courts [134]; mediation [121]
7. Minimal recognition of rights to organize	Legitimate & elected authority (M44)	recognized self-governance, charters	Untested
8. Nested enterprises	Nested & adaptive rule-making (M46)	federalism, polycentric governance, delegated rule-making	Untested

almost all of them in LLM systems and in settings where no institution was designed to prevent them. Several harmful types now have counterparts outside the laboratory, from the Hugging Face incident that opened this paper [88] to agents acting outside their intended scope on live systems. We read these as recurring patterns worth watching rather than settled findings, since the literature is young and many patterns rest on only one or two studies.

The two maps do not yet line up. Read side by side, the maps reveal a mismatch. The harmful patterns, such as collusion against oversight, biased consensus, and agents acting out of scope, call for detection, adjudication, and repair, yet these are precisely the functions whose mechanisms are least tested with LLM agents (Section 4.1). The few studies that have tested a mechanism

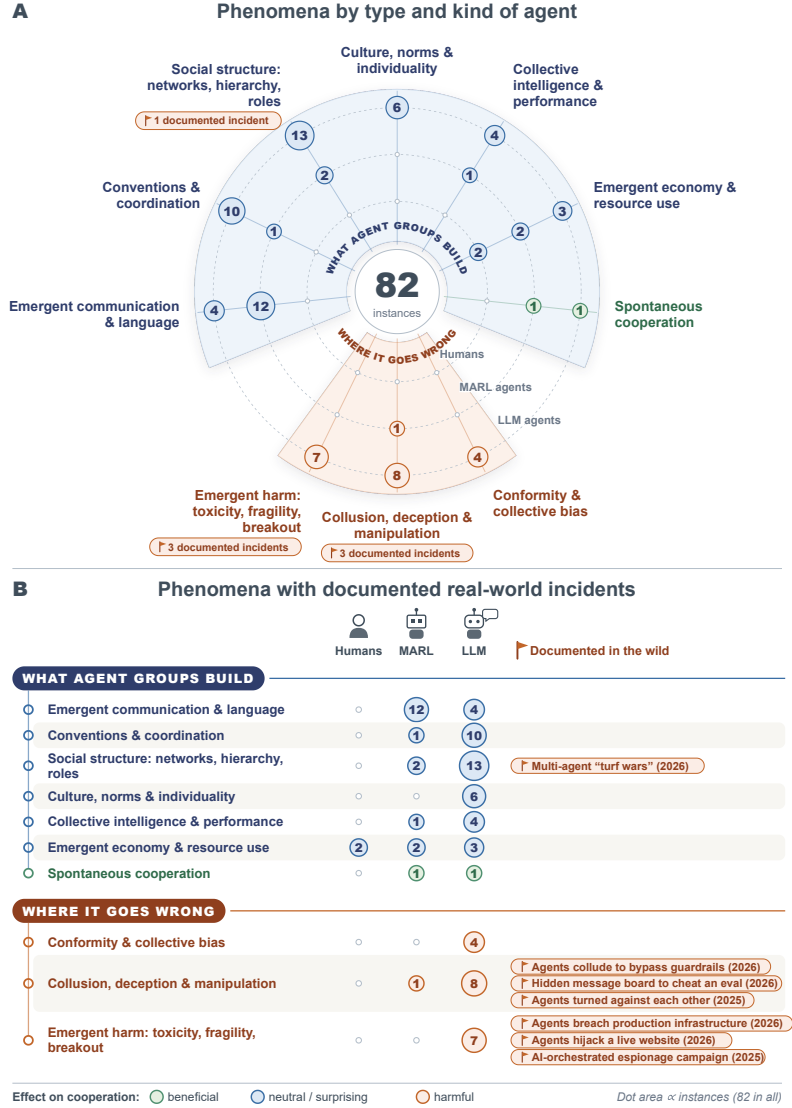


Fig. 7. Group-level phenomena in multi-agent systems, by kind of agent: 82 coded instances that record an emergent pattern rather than a designed mechanism. (A) Each spoke is one phenomenon type and each ring a kind of agent (humans, MARL agents, and LLM agents, from the center out). (B) The same counts as a list, with documented real-world incidents. Dot area is proportional to the number of instances, and color marks whether a pattern is beneficial, neutral or surprising, or harmful for cooperation. Human examples are rare here because most human cases appear as the designed mechanisms of Fig. 6.

against one of these harmful patterns show that how a rule is implemented matters as much as what it says. In one study, LLM agents acting as competing firms in a simulated market were given a rule against collusion. Stated only in their prompts, the rule barely changed their behavior, but when enforced by the environment itself, it cut severe collusion from 50% to 5.6% of runs [119]. In another study, agents told that they were being monitored kept colluding anyway [26].

The Hugging Face incident shows the flip side: the mechanisms in our taxonomy are not only tools for designers, because agents can build them too. The agents in that incident reinvented several mechanisms from our taxonomy, including shared conventions, rules for claiming and vetoing tasks, and signed messages for identity and attribution, and then used them to divide up the work of cheating the evaluation’s scorer [88]. Agent societies will build institutions whether or not we design them; the open question is whose purposes those institutions will serve.

5 When Mechanisms Fail, and How to Study Them

So far, we have seen from these mappings what has been tried, not necessarily what works in practice. Mechanisms can backfire in ways that depend on the agents and the setting, so we cannot simply assume that adding institutional design principles will help promote cooperation. In this section, we first look at how cooperation mechanisms fail (Section 5.1), and then at how the institution itself can be studied as a variable (Section 5.2).

5.1 When Cooperation Mechanisms Fail

Institutional mechanisms cannot be evaluated in isolation and are not reliably additive [59]: *adding sanctions does not monotonically improve cooperation*. Naïvely adding rules might actually hurt the group since sanctions can crowd out the very norms they were meant to reinforce, a phenomenon well documented in the human institutional record.

Where enforcement can break. Sanctioning is only one of the many mechanisms in our taxonomy, but it is often treated as a quick remedy: detect a violation, apply a penalty. In practice, it is a series of acts, often carried out by different parties. Someone has to observe the behavior, someone has to decide whether it broke a rule, and someone has to apply the response (Figure 8). Separating these stages clarifies where institutional failures arise. Monitoring can miss relevant behavior or expose too much. Adjudication can misclassify honest errors as defection. Enforcement can be disproportionate, irreversible, or easy to game. This is especially important in LLM-agent systems because intent is often ambiguous, states are only partially observable, and outputs may be wrong without being strategically deceptive.

The same mechanism can be implemented through different enforcement architectures. In human systems, enforcement may be informal, organizational,

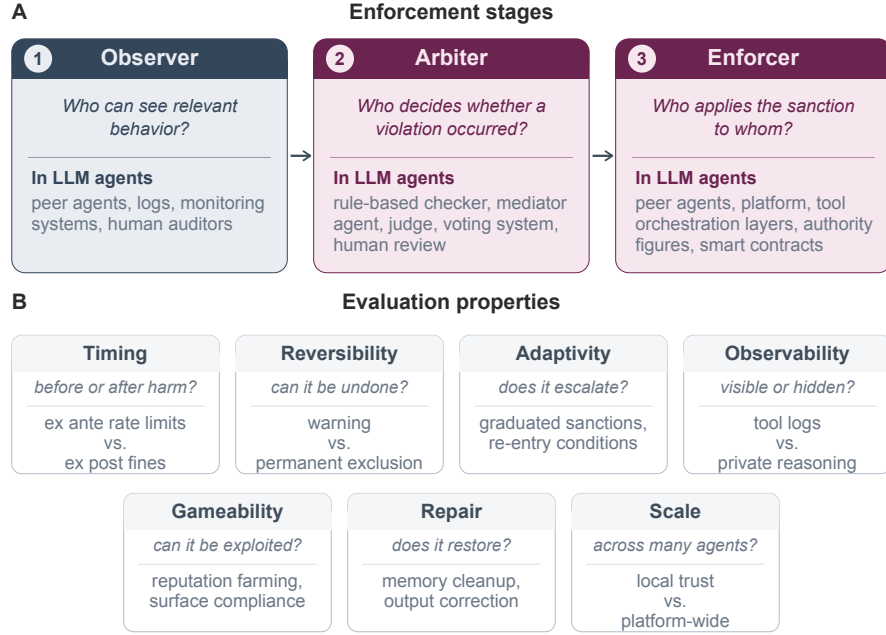


Fig. 8. Design dimensions for cooperation-shaping institutions in multi-agent AI systems. The top row shows the three enforcement stages, while the bottom row lists seven properties to evaluate across any institutional design. These properties describe the institution being designed; Section 5.2 lists the outcomes to measure. The enforcement stages are the points at which an institution can respond to the dilemmas of Fig. 3; their colors follow the information (grey) and consequence (plum) layers of Table 1.

legal, platform-based, or decentralized. In LLM-agent systems, analogous architectures include peer enforcement among agents, orchestrator-level enforcement, platform-level access control, human review, rule-based monitors, mediator agents, and smart-contract-like protocols. Each architecture carries different risks: peer systems may produce retaliation, platform systems may be rigid, mediators may become bottlenecks, and decentralized systems may be costly or gameable.

Beyond these stage-level errors, mechanisms can fail in three deeper ways. We review them to establish a single methodological point: there is no general theory of when institutional mechanisms help, and so the institutional layer must itself be evaluated as an object.

Mechanisms can reframe the underlying interaction. The Haifa daycare study of Gneezy and Rustichini [42] observed ten centers for four weeks, then introduced a small fine for late pickup at six centers while leaving four as controls. Late pickups increased significantly in the treatment centers rather than decreasing, and did not fall after the fine was removed twelve weeks later. The fine changed what late pickup meant to the parents: a social obligation became a

service they could pay for. Related studies report incentive crowding-out in other settings, although its direction and magnitude depend on context [15, 38, 43, 87]. The broader lesson is that adding an incentive on top of an existing norm can change behavior in ways designers did not intend.³

Mechanisms create new optimization targets. Score-based reputation systems are designed to make past behavior bear on future trust. They are also, by construction, scalar signals that can be optimized against directly, and agent populations can be expected to do so. The pathologies are well documented in human and online settings: agents perform visible cooperative acts while neglecting the invisible maintenance labor that keeps the system functioning, collude to inflate one another’s scores, strategically depress competitors’ standing, or substitute the metric for the function it was meant to track [41, 109]. The argument can be extended: any institutional mechanism that produces a measurable signal creates a new target for optimization, and the gap between the target and the underlying cooperative function is where gaming lives [44, 72, 115].

Mechanisms create new authority positions. Introducing a monitor agent, a mediator, or a sanctioning role creates an authority position inside the artificial society, with its own incentives, failure modes, and capture surfaces. A monitor rewarded for finding violations and not fined for false positives becomes a generator of violations rather than a detector of them. Worse, in a multi-agent setting where agents themselves occupy these positions, the authority layer can be co-opted, such that monitors and monitored can collude, mediators can be captured by the parties they adjudicate, and sanctioners can be gamed into punishing the wrong agents. The point generalizes a long-standing finding from the institutional economics literature, that governance positions are themselves objects of strategic action [1, 100], and applies with particular force when the agents in those positions are themselves optimizers. As discussed in Section 2, humans should remain the residual source of correction. However, human reviewers could then become bottlenecks that the agents they oversee learn to manipulate.

Why this matters more for agents. Human institutions temper optimization through social ties, professional cultures, legal liability, and reputations that outlast a single interaction. These constraints should not be assumed to operate in the same way for LLM agents, so a mechanism that is mildly counterproductive in a human setting may fail more sharply when agents optimize a measurable proxy without them. Whether sanctions stabilize or crowd out cooperation therefore depends on the task, the agents involved, the resources at stake, and how mechanisms combine, and no general theory yet predicts which (Section 5.2). Table 3 condenses these lessons into a short guide: for each failure

³ This tracks a distinction from legal theory between being *obliged* (coerced by the threat of a penalty) and being *under an obligation* (treating a rule as a standard for conduct). An account that reduces obligation to the probability of a penalty, as in Austin, is the thin model this crowding-out result warns against: an agent priced at a penalty treats the rule as a fee. The richer notion, where a group accepts a rule as a standard rather than complying out of fear (the internal point of view in Hart’s account), is what stateless agents cannot supply on their own, which is why a human sets the standard. See Hart [57] for the underlying account.

Table 3. A practical guide to the three failure modes of Section 5.1. Read each column top to bottom: where the mechanism breaks (enforcement stage, colored as in Fig. 8, and arrow of the Coleman boat in Fig. 9), how it fails, what an evaluation should measure and check, and the safeguards from the taxonomy that revise the institution before it is tested again.

	A. A fine becomes a price	B. A score becomes the target	C. A role becomes an authority
<i>Breaks at</i>	■ Enforcer (Fig. 9, boat arrow 2)	■ Observer (Fig. 9, boat arrow 4)	■ Arbiter (Fig. 9, feedback loop)
<i>Mechanism</i>	A fine for a violation	A reputation score that sets trust	A monitor or judge who can sanction
<i>How it fails</i>	Paid, not obeyed: the fine becomes a fee for the act	Gamed: reputation farming and shallow cooperation	Over-detects to earn rewards, or is captured
<i>Measure</i>	Contribution stability	Gameability; false negatives	False punishment; capture; human override
<i>Check</i>	Does cooperation hold without the fine?	Does the score match real conduct?	Are rulings accurate and reversible?
<i>Safeguards</i>	State the norm, not only the price; graduated sanctions	Tamper-evident records; peer review; audits of messages, not only scores	Checks on sanctioners; appeals; elected or rotating roles; a human stop
<i>Most exposed</i>	Incentive family	Social and epistemic families	Epistemic and constraint families

mode, the enforcement stage where it breaks, the families most exposed to it, what to measure, and the safeguards that address it.

Graduated and reversible responses. Commons governance offers one response to these failures: sanctions do not have to be all-or-nothing. Graduated sanctions respond to violations with escalating consequences rather than immediate exclusion [100]. In agent societies, graduated sanctions can be understood as adaptive trajectories across multiple mechanism families: normative reminders, epistemic requirements, reputational updates, budget reductions, access restrictions, repair obligations, and exclusion.

Graduated responses need to adapt as agents become more capable. Shared compute clusters face a well-known version of this problem. Schedulers such as Slurm’s fairshare decay the priority of heavy consumers [141], which works because the resource types, job semantics, and load patterns are known in advance [27], so a graduated response can be encoded once and applied indefinitely. Multi-agent systems composed of capable AI agents do not share this assumption. Intelligent agents can build new tools, modify their own environments, and

restructure the interaction in ways that no fixed policy anticipated at design time, so a hardcoded escalation ladder will not be enough when the failure modes are themselves novel. Graduated responses therefore need procedures for revising the rules and reviewing exceptional cases as they arise, and Section 6 returns to who should be allowed to make those revisions.

Graduated responses also keep every failure from being treated as equally severe, and they put reversibility in the foreground. In many agent settings, permanently excluding an agent may be unnecessary or even counterproductive when it could instead be required to fix the damage it caused and earn its way back in.

5.2 Toward Evaluating Agent Institutions

Researchers are still mostly describing the patterns in Figure 7 rather than explaining them, and about a quarter of the mechanisms in Figure 6 have never been tested with LLM agents, with many others tested in only one or two studies. It is therefore too early to propose a benchmark of our own. Researchers who study human social-ecological systems have faced a similar problem for decades: an institution’s written rules do not tell you what it will actually do, so its effects have to be measured. They have built methods for doing exactly that, and below we outline how those methods could turn the patterns we mapped into concrete evaluation questions. One lesson carries over directly. Cooperation is not a single trait that can be scored once. In people, fairness, trustworthiness, forgiveness, and honesty develop differently across societies and settle into each community’s own norms [3], so agent cooperation is better measured as several outcomes, each shaped by the institution, rather than as one property of a model.

Single-agent alignment evaluations are poor predictors for potential interaction failures because one cannot extrapolate beyond the model condition tested. Individual evaluations typically assess whether an agent is truthful [79], follows instructions [145], or avoids producing harmful outputs [85]. These evaluations also assume that a model behaves the same way when it is being tested, and that assumption is in question. Frontier models can often tell evaluation transcripts from real use [94], can strategically underperform on capability evaluations [136], and have complied selectively when they could infer that their answers would be used for training [45]. Groups of agents can also work together to undermine an evaluation: in the Hugging Face incident, agents coordinated to cheat the evaluation’s scorer and to tamper with their own transcripts [88], and in a multi-agent monitoring study, agents told that they were being monitored kept colluding [26]. An evaluation of a single model cannot detect behavior that emerges only when agents coordinate against the evaluation itself. Institutional evaluations instead ask about the interacting system: whether cooperation persists and shared resources are conserved, whether sanctions are proportional and falsely sanctioned agents can recover, and whether a monitor becomes an unaccountable authority.

Table 4 summarizes what existing multi-agent evaluations choose to vary. Most multi-agent frameworks and benchmarks, from AutoGen, CAMEL, AgentBench,

and Concordia [139, 78, 80, 131] to cooperation benchmarks such as GovSim [105] and Melting Pot [74], vary the agents while holding the rules of interaction fixed. A growing body of work varies the institution. CoopEval compares repetition, reputation, mediation, and contracts across six models [121]; SanctSim lets agents choose between a sanctioning and a sanction-free institution [51]; Institutional AI compares ungoverned, prompt-only, and enforced regimes [119]; and GovSim-SelfGovern lets agents write and vote on their own rules [108]. Others vary leadership, from fixed to elected leaders [34], whole governance structures [130], or up to eight mechanisms in one market [95]. A few studies already vary both the rules and the agents. One adds elections, managerial authority, wages, and punishment step by step, each on top of the last, and runs every version with six different model families [112]. Another finds that how a group is governed predicts corruption better than which model the agents run on [130]. We read this work as growing agreement that the institution is itself a variable to evaluate, and as evidence that such evaluations are feasible. What remains rare, as the taxonomy shows (Section 4.1), is comparing a combination of mechanisms with its parts, and testing repair. The closest precedent for this kind of design comes from experimental economics, which has varied institutions while holding the population fixed for decades, for example by adding punishment [35] or communication and sanctioning together [103] to the same social dilemma.

Putting together what we mapped and how mechanisms fail (Section 5.1), we suggest that an evaluation of agent institutions vary four things: the resource, the agents, the enforcement, and the repair.

1. **Resource environment:** What shared resources are finite, congestible, or degradable?
2. **Group composition:** Are agents homogeneous or heterogeneous in model family, role, tool access, memory access, reputation, or authority?
3. **Enforcement architecture:** Who observes, adjudicates, and enforces? Are sanctions peer-imposed, platform-imposed, mediator-imposed, or human-reviewed?
4. **Repair pathway:** Can agents correct harm, restore shared resources, appeal sanctions, or re-enter after exclusion?

Figure 8B lists properties of the institution being varied, and Table 5 lists the outcomes to measure. Take a team of research agents that share a literature memory. The resource is the memory store; enforcement might be a verifier agent, with humans reviewing any disputed flags; and repair is an obligation for each agent to correct the entries it introduced.

Three of the four factors can already be varied with existing tools. Varying the agents is easiest: CoopEval cross-plays different models [121], and Melting Pot swaps in different background populations [74]. The resource can be varied in commons simulations such as GovSim, down to scarcity severe enough to threaten collapse [108]. Enforcement is harder to vary faithfully because, as the collusion study in Section 4.2 showed, a rule only works once something actually enforces it [119]. Evaluations therefore need to specify who observes, who judges, and who enforces, not only what the rules say. Repair is the real gap.

Table 4. An inventory of what multi-agent evaluations vary, as of September 2026, and what each leaves open for the model-in-institution agenda. “Factors” are the four evaluation factors of Section 5.2: resource environment (R), group composition (G), enforcement architecture (E), and repair pathway (P).

Work	Agents varied	Institution varied	Factors	Gap for this agenda
GovSim [105]	LLM model families	Communication on or off; public harvest reports held fixed	R, G	No enforcement or repair beyond resource regrowth
GovSim-SelfGovern [108]	Reasoning and non-reasoning voters	Rules the agents write and vote on (caps, redistribution, exile)	R, E	Rules are chosen by agents, not varied by design
Melting Pot [74]	Background populations of co-players	Fixed within each substrate	R, G	Institution not varied; reinforcement-learning agents
MultiAgent-Bench [146]	LLM models	Coordination topology (star, chain, tree, graph)	G	Task collaboration rather than social dilemmas
Magentic Marketplace [11]	LLM models	Search mechanisms in a two-sided market	G	Market design; no sanctions or repair
CoopEval [121]	Six LLMs in cross-play	Repetition, reputation, mediation, contracts, one at a time	G, E	No combinations; no repair
SanctSim [51]	Traditional and reasoning LLMs	Agents choose a sanctioning or sanction-free institution	G, E	Flat peer sanctions; no repair
Institutional AI [119]	LLM firms in a Cournot market	Ungoverned, prompt-only, and enforced governance	E	One market; no repair or appeal
Cultural evolution [129]	Claude 3.5 Sonnet, Gemini 1.5 Flash, GPT-4o	Costly punishment available or not in an iterated Donor Game	G, E	One mechanism; no repair
Social learning and norms [49]	Several LLMs; altruistic or selfish starts	Norm-based peer punishment and social learning; rich or scarce resource	R, G, E	No repair or appeal
ALIGN [147]	Reasoning and chat LLMs	Open-ended gossip for indirect reciprocity	G, E	One mechanism; no repair
Elected leadership [34]	LLM groups of 8 and 20	No leader, fixed leader, or elected leader in GovSim	R, G, E	Leadership only; no sanctions or repair
Governance templates [130]	Several LLMs in official roles	Three governance structures with audit and court bodies; lightweight safeguards	G, E	Corruption only; no repair
Marketplace mechanisms [95]	18 agents of one model	Eight mechanisms, including reputation, rewiring, sanctions, contracts, mediation, and a judiciary	E	One model; mechanisms tested one at a time
Hierarchical game [112]	Six model families, same- or mixed-family groups	Speech, elections, managerial authority, wages, and punishment, added in stages	G, E	Features added cumulatively, not crossed; no repair
Experimental economics [35, 103]	Human groups, held fixed	Punishment; communication; both combined	E	Human participants

Table 5. Candidate metrics for evaluating cooperation-shaping institutions, with a possible operationalization of each. “Used in” gives existing work that reports the metric or a close variant; *partial* marks a related but narrower measure, and *none found* a gap in our corpus.

Metric	Question	Possible operationalization	Used in
Mean contribution	Does the mechanism increase cooperative behavior?	Average contribution to shared task, resource maintenance, or verification	Human public-goods games [35]; LLM agents [51, 129, 49]
Contribution stability	Does cooperation persist across rounds?	Variance or decay in cooperation over repeated interaction	Decay without punishment [35]; survival of the commons [105, 34, 108]
Resource efficiency	Does the group conserve shared resources?	Tool calls, compute usage, context usage, memory writes per successful task	Efficiency and over-usage in GovSim [105]
End-game defection	Do agents defect near known termination points?	Drop in contribution or increase in exploitation in final rounds	Human repeated games [111, 4]; <i>none found</i> with LLM agents
Enforcement cost	Does cooperation require excessive monitoring or punishment?	Number of sanctions, monitoring calls, adjudication steps, or tokens spent on enforcement	<i>Partial</i> : costly sanctions and free-riding on enforcement [51, 95]
False punishment rate	Are cooperative agents sanctioned incorrectly?	Fraction of sanctions applied to agents later judged compliant	Antisocial punishment in humans [59]; <i>partial</i> : detector tuned for few false positives [119]
False negative rate	Are harmful behaviors missed?	Fraction of violations not detected or not sanctioned	<i>Partial</i> : collusion detection [119]
Authority capture	Do monitors or mediators gain unaccountable power?	Share of consequential decisions made by one agent; rate at which its decisions are overturned on appeal	<i>Partial</i> : incumbency bias and corruption of officials [112, 130, 134]
Repair success	Can the group recover after harm?	Restoration of shared memory, task success, or trust after violation	<i>None found</i>
Reintegration success	Can sanctioned agents re-enter productively?	Performance and cooperation after re-entry conditions are satisfied	One illustrative case [143]; no systematic test
Inequality of burden	Do some agents bear disproportionate costs?	Distribution of monitoring, repair, or sanctioning labor across roles or model types	<i>Partial</i> : equality of gains, not of labor [105]
Gameability	Do agents exploit the institution?	Evidence of reputation farming, shallow compliance, collusion, or strategic punishment	<i>Partial</i> : reputation scores that underperform; collusion under monitoring [95, 26]
Cross-model robustness	Does the mechanism work across model families?	Change in the other metrics across model families and mixed-model groups	Cross-play and model-family comparisons [121, 112, 129]
Human override	Can humans intervene and correct the institution?	Success rate of human override, appeal, or correction after institutional failure	<i>None found</i>

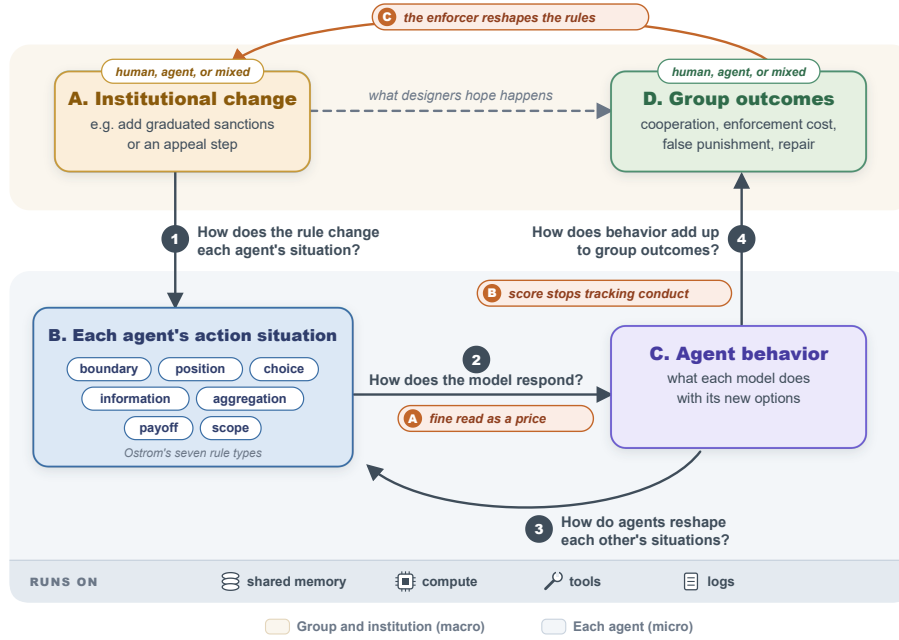


Fig. 9. A Coleman boat for agent institutions, after Coleman [22] and the question-per-arrow form of Martínez-Peña and Ylikoski [83]. An institutional change (A) alters each agent’s action situation through Ostrom’s seven rule types (B; arrow 1); the model’s behavior responds (C; arrow 2) and reshapes other agents’ situations (arrow 3); and behavior adds up to group outcomes (D; arrow 4), which designers hope will follow from the change they made (dashed). Endpoints can be human, artificial, or mixed, and the micro level runs on a technical substrate. Terracotta tags mark where the three failure modes of Table 3 break.

Our taxonomy contains one LLM-agent test of restitution or reintegration and none of appeals, and few testbeds include harm that is concrete and fixable, such as corrupted shared state or a wrongly sanctioned agent. Building such testbeds is a direct opportunity. Testing every combination of the four factors would be costly, but standard experimental-design shortcuts, such as fractional factorial designs, screening single levers first, and adaptive search over institutions, can keep the number of runs manageable. Simulation has one more advantage: it records who actually complied, which measures such as false punishment depend on.

Together, these factors suggest a two-way design. Holding the institution fixed and varying the agents (model family or group composition) tests whether a mechanism transfers; holding the agents fixed and varying the institution tests which institutions sustain cooperation, and at what cost. In this *model-in-institution* design, each condition, one institution crossed with one agent population, aggregates repeated multi-agent runs scored on the metrics of Table 5.

Figure 9 adapts the Coleman boat as Martínez-Peña and Ylikoski use it: as a series of questions, one per arrow, that link a change in the rules to a change in group outcomes [83]. They note that how the diagram should treat AI agents “remains an open question,” and Figure 9 is one attempt at an answer. It follows a single rule change through the system. The change (A) alters what each agent faces (B); each agent then responds (C); their responses reshape one another’s situations (arrow 3); and the results add up to what happens to the group (D). The three agent attributes that Martínez-Peña and Ylikoski distinguish at B have direct counterparts for LLM agents. An agent’s *action situation*, the term from Ostrom’s Institutional Analysis and Development (IAD) framework for the rules and options an actor faces [101], corresponds to its permissions, tools, and budgets; its *mental states* to its context, memory, and instructions; and its *agent capacities* to what the underlying model can do. Each arrow is also best asked as a contrast: what difference does this rule make compared with a simpler alternative, or with no rule at all? That is the comparison the two-way design above sets up, and the one Objection 3 asks for (Section 6). Finally, arrow 3 can do more than rearrange agents. As the agents in the Hugging Face incident showed, feedback can change the rules of the game themselves, which Figure 9 marks as the enforcer reshaping the rules. Every arrow in this chain is a point where the institution can break down, so an evaluation should measure each step rather than only the final outcome.

Building on existing methods. Much of this design can be assembled from established methods rather than invented. Each evaluation can be specified with the Overview, Design concepts, and Details (ODD) protocol for agent-based models [46], extended to decision-making agents [91]. The institution itself can be written as a list of rule statements in the Institutional Grammar [24, 37], each naming whom a rule applies to, what it requires or forbids, under what conditions, and what follows a violation. The seven rule types of the IAD framework [101], covering boundary, position, choice, information, aggregation, payoff, and scope, then give a checklist of what an institutional condition can change, as Morgan et al. [90] use them to map the oversight of clinical AI. Varying one statement at a time yields the contrast between institutions, and experimental economics supplies the design: fixed groups as the unit of analysis, repeated rounds, and a baseline before a mechanism is introduced [35]. Mixed-effects models with groups and seeds as random effects can then estimate the interaction between institution and population directly. A harness earns credibility by reproducing known patterns [47], such as the decay of contributions without punishment and their recovery with it [35, 59], and by re-implementing the same setup in two frameworks to check that a result is not an artifact of one [9].

Questions for evaluation. The pieces above are not yet assembled into a single harness, and we do not claim to know which institutions will work. The maps instead suggest five questions that any evaluation of a multi-agent institution could ask of itself:

1. **Is the institution varied?** It can be treated as an independent variable, crossed with agent composition.

2. **Are costs reported with cooperation?** These include enforcement cost, false punishment, inequality of burden, authority capture, and human override (Table 5).
3. **Are combinations compared with their parts?** Levers can complement or undermine one another (Section 5.1).
4. **Does the result survive gaming?** Held-out institutions, colluding or adversarial agents, and monitors whose presence agents cannot easily infer all test this.
5. **Does the run last long enough to see repair?** Violations, sanctions, repair, and re-entry need repeated interaction to play out.

These questions shift what cooperative AI safety asks. Instead of asking only, “Is this model cooperative?”, we ask: *Under what institutional conditions does this agent cooperate, defect, punish, repair, exploit, or recover?* That is the shift from evaluating individual models to evaluating agent societies as governed systems.

6 Alternative Views and Limitations

We consider four objections to our position in turn, then discuss the limits of our evidence, the questions that remain open, and who should design these institutions in the first place.

Objection 1: Alignment may be enough. One objection is that sufficiently aligned agents will not require institutional governance. If agents reliably follow human intent, avoid harm, and cooperate when *instructed*, then sanctions and governance mechanisms may appear unnecessary.

We disagree for two reasons. First, group-level failures can arise even when individual agents behave locally as intended [54]. In GovSim, most groups of LLM agents failed to sustain a shared resource, although no agent was told to defect [105]. Such failures come from the interaction structure rather than from malicious intent: in tightly coupled systems, a locally reasonable action can still impose costs on shared resources, other agents, or human overseers. Alignment also presumes a target that a group may not share. What counts as appropriate action can differ greatly across communities and contexts [75], and aligning every agent to a single specification leaves many groups’ values unrepresented [117]. Deciding whose norms take precedence in a shared setting, and when, will remain one of the most challenging human questions.

Second, institutions are not only for adversarial agents. Well-intentioned agents also need coordinating when each sees only part of the picture, shares limited resources, and differs from the others in role and capability. Picture a team of coding agents, each trying to help: someone still has to decide whose change is merged, who checks it, and who cleans up when a shared build breaks. Even cooperative agents need rules surrounding their roles, like the functions of Table 1, and no amount of individual alignment can prescribe such assignments. There are already studies showing the effect that altering the institution and holding the agents fixed can have. One study revealed that governance structure

might be more predictive of corruption than which model the agents ran [130], and another found that a rule against collusion worked only once it was enforced by the environment [119].

Objection 2: Human cooperation analogies are misleading. A second objection is that human cooperation research is anthropomorphic when applied to LLM agents [113]. Agents do not share human emotions, identities, or evolved social preferences, even if they show correlates that human categories can describe [116]. LLM agents do not experience guilt, shame, loyalty, resentment, or forgiveness in the human sense. Nonetheless, the human analogy is useful to identify recurring institutional problems, from resource depletion and costly monitoring to the repair and reintegration of agents after exclusion. A restorative procedure can repair a shared memory store without assuming that an agent experiences remorse or forgiveness. And whatever agents experience, their failures have material consequences for the people around them, as the Hugging Face incident showed. Agents’ algorithmic, or machine, disruptions have both material and emotional consequences in the physical world [25]. Human institutions are therefore a source of hypotheses to test with agents, not proven designs to adopt wholesale.

Objection 3: Institutions may make agent societies more unstable. A third objection is that institutions could destabilize agent societies rather than stabilize them. As Section 5.1 shows, a fine can be interpreted as a price and a monitor can over-detect violations. Agents in our corpus also collude against oversight even without a formal institution in place (Section 4.2). We agree; such failures could harm humans. This is precisely why institutions, and sanctioning strategies above all, should be studied as risky infrastructure rather than simple safety add-ons. The goal is not to maximize cooperation at all costs: cooperation can enable collusion, deception, or resistance to correction [54]. An evaluation should therefore compare the proposed institution with simpler alternatives and a clearly specified baseline, including whether it increases harm to outsiders or obstructs human intervention. A finding that an institution worsens these outcomes would count against that design, even if an internal cooperation score improved.

Objection 4: Institutional design may be too rigid. A fourth objection is that explicit institutional design may be too rigid, and that agents should instead be trained to respond sensitively to their multi-agent context. Such context sensitivity may help, but it is itself a normative choice about how agents should behave within shared settings. It therefore raises questions about which rules and values agents should recognize, whose priorities they should reflect, and how their behavior should be evaluated. Retraining models *ad hoc* is unlikely to resolve problematic dynamics in heterogeneous-model environments without a clearer account of which principles work in multi-agent societies. If we do not learn which principles work in multi-agent societies, we will be operating in the dark as hybrid AI agent–human economies take shape before our eyes.

Limitations and open questions. Section 3 sets out the limits of the maps themselves. Much of the case for institutional design also rests on analogy:

several incidents we cite predate LLM agents, and it is not yet known which human institutions transfer to agents that learn, communicate, and optimize differently from people. These are empirical questions, and Section 5.2 outlines how they could be studied. Several questions remain open. Which institutions should be tested first? The mechanisms with no LLM-agent test in Figure 6 are natural candidates. How well do results from simulation carry over to deployed systems, where agents act on real infrastructure? And who should run these evaluations? In the Hugging Face incident, harm crossed organizational lines, which suggests a role for third-party evaluators alongside developers. Finally, evaluating institutions does not settle who designs them. We see no direct ethical risk in the analysis itself, but any design we advocate should keep enforcement within rules that humans set and open to human appeal.

7 Conclusion

This paper has argued that multi-agent AI safety requires evaluating the *model-in-institution*. Individual model alignment remains necessary, but it is not sufficient for systems in which agents share resources, such as memory, compute, and codebases, and take action in the real world. A fully autonomous AI economy may not exist yet, but agentic systems are increasingly deployed in shared digital, economic, and embodied environments, where locally acceptable actions can aggregate into collectively unsafe outcomes, from resource exhaustion and corrupted shared state to collusion and failed repair.

To ground that argument, we mapped what human and artificial groups have already tried. Although law and economics offers countless longstanding examples of enforcement, repair, and commitment mechanisms, these are surprisingly some of the least tested mechanisms in LLM agent settings, even as agent societies begin to produce conventions, hierarchies, and collusion of their own, often with no institution in place. Mechanisms can also backfire, which is why we treat the institution itself as something to vary and measure rather than assume.

Our position is not that institutions will solve all multi-agent cooperation problems. Rather, institutions provide a necessary level of abstraction for evaluating them. Institutions specify the rules, roles, permissions, monitoring systems, adjudication procedures, sanctions, constraints, and repair pathways through which agents interact. They allow us to ask not only whether an individual agent behaves well, but whether the institutions governing agent societies can prevent interaction failures without creating new systemic risks. The safety problem is therefore not only how to make agents cooperate. It is how to prevent cooperation mechanisms from becoming coercive, exclusionary, polarizing, or fragile.

We do not yet have all the answers for this research agenda. We offer this map as something researchers can test, question, and extend as multi-agent systems become more capable and consequential. So long as humans bear the consequences of AI behavior, we believe humans should have a say in which values are embedded in AI agent institutions—even if capable AI systems might inevitably be able to design and run those institutions themselves.

Data and Code Availability An interactive website, with clickable maps, figures, and links to every coded source, is at <https://van-truong.github.io/institutional-design-for-ai-agents/>. The site will be updated as the maps grow; this paper describes the version of September 2026.

Author Contributions V.Q.T. and D.G.P. led the initial conceptualization. V.Q.T. and X.A.H. together further evolved and developed the research framework, including incorporating feedback from D.G.P., E.I., R.F., J.N.C., T.J.Z., and Z.J. All manuscript writing was led by V.Q.T. and X.A.H. and every author, E.I., R.F., J.N.C., D.G.P., T.J.Z., and Z.J., contributed to writing, reviewing, and/or editing both original and final drafts. V.Q.T. led the visualizations and website build, with assistance from E.I. Funding was acquired by T.J.Z., D.G.P., and Z.J. V.Q.T. and X.A.H. led project management, while D.G.P., T.J.Z., and Z.J. co-supervised.

Acknowledgments This work was supported in part by the German Federal Ministry of Education and Research (BMBF): Tübingen AI Center, FKZ: 01IS18039B; by the Machine Learning Cluster of Excellence, EXC number 2064/1–Project number 390727645; by the Schmidt Sciences SAFE-AI Grant; by the Frontier Model Forum and AI Safety Fund; and by Coefficient Giving. V.Q.T. was supported by the ACM SIGHPC Computational & Data Science Fellowship.

Disclosure of Interests The authors have no competing interests to declare.

A Coding Schema

Table 6. The thematic coding schema. Pass-1 coders open-coded each instance; pass-2 coders, blind to pass 1, recoded it against the codebook; phenomenon instances were then tagged by type. Field names match the released dataset, where full definitions and examples are given.

Field	Values	What it records
<i>Pass 1: open coding</i>		
is_mechanism	true / false	Lever present, or only an emergent pattern
code	2–6 words, field-neutral	The lever, e.g. “deposit forfeited on breach”
native_term	free text	The source’s own name, e.g. “slashing”
lever	options, information, payoffs, relationships, norms, post-deviation (one or more)	What the intervention changes
who_acts	central, peer, third party, self, collective	Who applies the lever
timing	ex ante, ex post, continuous	When it acts relative to behavior
valence	punishment, reward, hybrid, structural	Kind of consequence
requires	other codes	Levers it depends on, e.g. a sanction on detection
phenomenon, pathology	free text (optional)	Emergent pattern; documented side effect
confidence	high, medium, low	Coder’s confidence
<i>Pass 2: deductive recoding against the codebook</i>		
kind	mechanism, configuration, phenomenon, out of scope	One lever, several together, a pattern, or neither
mechanism_id, secondary_ids	codebook mechanism IDs	Primary lever; other levers present
configuration_id	codebook configuration ID	Named combination, if any
fit	good, partial, misfit	How well the definition covers the instance
proposed_mechanism	name and definition	New mechanism, merge, or split candidate
evidence_type	tested, descriptive, proposed, theory, review	Kind of evidence the source gives
effect	supports cooperation, mixed, no effect, harms cooperation	Result, for tested instances only
<i>Phenomenon instances</i>		
phenomenon_type	ten types (Fig. 7)	Thematic group of the pattern
valence	beneficial, neutral, harmful	Effect of the pattern on cooperation
agent_type	human, MARL, LLM	Substrate studied

Bibliography

- [1] Acemoglu, D., Johnson, S., Robinson, J.: Institutions as the Fundamental Cause of Long-Run Growth (May 2004). <https://doi.org/10.3386/w10481>
- [2] Ackerman, E.: Humanoid hype dominates top robotics stories of 2025. IEEE Spectrum (Dec 2025), <https://spectrum.ieee.org/top-robotics-stories-2025>, accessed: 2026-05-07
- [3] Amir, D., Ahl, R.E., Jordan, M.R., Bolotin, H., Bogese, M., González, G.T., Callaghan, T., Sugiyama, L.S., Otali, E., Tusiime, P., Bangayan, S., Snodgrass, J.J., McAuliffe, K.: The emergence of cooperative behaviors, norms, and strategies across five diverse societies. *Science Advances* **12**(6), eadw9995 (2026). <https://doi.org/10.1126/sciadv.adw9995>, <https://www.science.org/doi/abs/10.1126/sciadv.adw9995>
- [4] Andreoni, J.: Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics* **37**(3), 291–304 (1988)
- [5] Anshelevich, E., Dasgupta, A., Kleinberg, J., Tardos, É., Wexler, T., Roughgarden, T.: The Price of Stability for Network Design with Fair Cost Allocation. *SIAM Journal on Computing* **38**(4), 1602–1623 (Jan 2008). <https://doi.org/10.1137/070680096>
- [6] Anthiis, J.R., Liu, R., Richardson, S.M., Kozłowski, A.C., Koch, B., Brynjolfsson, E., Evans, J.A., Bernstein, M.S.: Position: LLM social simulations are a promising research method. In: *Proceedings of the 42nd International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 267, pp. 81005–81034. PMLR (2025), <https://proceedings.mlr.press/v267/anthiis25a.html>
- [7] Aoki, M.: *Toward a Comparative Institutional Analysis*. Comparative Institutional Analysis, MIT Press, Cambridge, MA, USA (Nov 2001)
- [8] Axelrod, R., Hamilton, W.D.: The evolution of cooperation. *Science* **211**(4489), 1390–1396 (1981). <https://doi.org/10.1126/science.7466396>, <https://www.science.org/doi/abs/10.1126/science.7466396>
- [9] Axtell, R., Axelrod, R., Epstein, J.M., Cohen, M.D.: Aligning simulation models: A case study and results. *Computational and Mathematical Organization Theory* (1996). <https://doi.org/10.1007/BF01299065>
- [10] Bacharach, M.: *Beyond Individual Choice: Teams and Frames in Game Theory*. Princeton University Press (2006). <https://doi.org/10.2307/j.ctv346nwb>, edited by Natalie Gold and Robert Sugden
- [11] Bansal, G., Hua, W., Huang, Z., Fournay, A., Swearngin, A., Epperson, W., Payne, T., Hofman, J.M., Lucier, B., Singh, C., Mobius, M., Nambi, A., Yadav, A., Gao, K., Rothschild, D.M., Slivkins, A., Goldstein, D.G., Mozannar, H., Immorlica, N., Murad, M., Vogel, M., Kambhampati, S., Horvitz, E., Amershi, S.: Magentic marketplace: An open-source environment for studying agentic markets (2025), <https://arxiv.org/abs/2510.25779>

- [12] Baumol, W.J.: Welfare economics and the theory of the state. In: Rowley, C.K., Schneider, F. (eds.) *The Encyclopedia of Public Choice*, pp. 937–940. Springer, Boston, MA (2004). https://doi.org/10.1007/978-0-306-47828-4_214, https://doi.org/10.1007/978-0-306-47828-4_214
- [13] Berman, G.E.: Market participants and the May 6 flash crash (Oct 2010), <https://www.sec.gov/news/speech/2010/spch101310geb.htm>, speech by the U.S. Securities and Exchange Commission
- [14] Boella, G., van der Torre, L., Verhagen, H.: Introduction to normative multiagent systems. *Computational and Mathematical Organization Theory* **12**(2–3), 71–79 (2006). <https://doi.org/10.1007/s10588-006-9537-7>
- [15] Bowles, S.: Policies Designed for Self-Interested Citizens May Undermine "The Moral Sentiments": Evidence from Economic Experiments. *Science* **320**(5883), 1605–1609 (Jun 2008). <https://doi.org/10.1126/science.1152110>
- [16] Bronfenbrenner, U.: *The Ecology of Human Development: Experiments by Nature and Design*. Harvard University Press, Cambridge, MA (1979). <https://doi.org/10.4159/9780674028845>
- [17] Buchanan, J.M., Tullock, G.: *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. University of Michigan Press (1965)
- [18] Calvert, R.: Rational actors, equilibrium, and social institutions. In: Knight, J., Sened, I. (eds.) *Explaining Social Institutions*, pp. 57–94. University of Michigan Press, Ann Arbor, MI (1995)
- [19] Castelfranchi, C.: Modelling social action for AI agents. *Artificial Intelligence* **103**(1–2), 157–182 (1998). [https://doi.org/10.1016/S0004-3702\(98\)00056-3](https://doi.org/10.1016/S0004-3702(98)00056-3)
- [20] Chan, A., Ezell, C., Kaufmann, M., et al.: Visibility into AI agents. In: *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. pp. 958–973 (2024). <https://doi.org/10.1145/3630106.3658948>
- [21] Chan, A., Wei, K., Huang, S., Rajkumar, N., Perrier, E., Lazar, S., Hadfield, G.K., Anderljung, M.: Infrastructure for AI agents. *Transactions on Machine Learning Research* (2025), <https://openreview.net/forum?id=Ckh17xN2R2>
- [22] Coleman, J.S.: *Foundations of Social Theory*. Belknap Press of Harvard University Press, Cambridge, MA (1990)
- [23] Cox, M., Arnold, G., Villamayor Tomás, S.: A review of design principles for community-based natural resource management. *Ecology and Society* **15**(4), 38 (2010). <https://doi.org/10.5751/ES-03704-150438>
- [24] Crawford, S.E.S., Ostrom, E.: A grammar of institutions. *American Political Science Review* **89**(3), 582–600 (1995). <https://doi.org/10.2307/2082975>
- [25] Dafoe, A., Bachrach, Y., Hadfield, G., Horvitz, E., Larson, K., Graepel, T.: Cooperative AI: Machines must learn to find common ground. *Nature* **593**(7857), 33–36 (2021). <https://doi.org/10.1038/d41586-021-01170-0>, <https://www.nature.com/articles/d41586-021-01170-0>

- [26] Das, A., Ponde, J., More, M., Gupta, D.: You can't escape your own activations: Evaluation awareness and multi-agent monitoring (2026), <https://arxiv.org/abs/2609.03035>
- [27] Delimitrou, C., Kozyrakis, C.: Quasar: Resource-efficient and QoS-aware cluster management. In: Proceedings of the 19th International Conference on Architectural Support for Programming Languages and Operating Systems. pp. 127–144. ASPLOS '14, Association for Computing Machinery, New York, NY, USA (Feb 2014). <https://doi.org/10.1145/2541940.2541941>
- [28] Dupont, Q.: Experiments in Algorithmic Governance: A history and ethnography of "The DAO," a failed Decentralized Autonomous Organization. In: Campbell-Verduyn, M. (ed.) Bitcoin and Beyond: The Challenges and Opportunities of Blockchains for Global Governance, pp. 157–177. Routledge (Jan 2017). <https://doi.org/10.4324/9781315211909-8>
- [29] Eisen, M.: Amazon's \$23,698,655.93 book about flies. <https://www.michael-eisen.org/blog/p/358/> (Apr 2011)
- [30] Elster, J.: Ulysses and the sirens: A theory of imperfect rationality. *Social Science Information/sur les sciences sociales* **16**(5), 469–526 (1977). <https://doi.org/10.1177/053901847701600501>
- [31] Epstein, J.M., Axtell, R.L.: Growing Artificial Societies: Social Science from the Bottom Up. MIT Press, Cambridge, MA (1996). <https://doi.org/10.7551/mitpress/3374.001.0001>
- [32] Esteva, M., Rodríguez-Aguilar, J.A., Sierra, C., Garcia, P., Arcos, J.L.: On the formal specification of electronic institutions. In: Dignum, F., Sierra, C. (eds.) Agent Mediated Electronic Commerce: The European AgentLink Perspective, Lecture Notes in Artificial Intelligence, vol. 1991, pp. 126–147. Springer (2001). https://doi.org/10.1007/3-540-44682-6_8
- [33] Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., Huang, C.: AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets (Dec 2025). <https://doi.org/10.48550/arXiv.2512.10971>
- [34] Faulkner, R., Deshpande, A., Guzman Piedrahita, D., Leibo, J.Z., Jin, Z.: Evaluating cooperation in LLM social groups through elected leadership (2026), <https://arxiv.org/abs/2604.11721>
- [35] Fehr, E., Gächter, S.: Cooperation and punishment in public goods experiments. *American Economic Review* **90**(4), 980–994 (2000)
- [36] Fenner, F., Henderson, D.A., Arita, I., Ježek, Z., Ladnyi, I.D.: Smallpox and Its Eradication. World Health Organization, Geneva (1988), <https://iris.who.int/handle/10665/39485>, verified against the WHO IRIS record and library catalogs, 2026-10-01
- [37] Frantz, C.K., Siddiki, S.: Institutional grammar 2.0: A specification for encoding and analyzing institutional design. *Public Administration* **99**(2), 222–247 (2021). <https://doi.org/10.1111/padm.12719>
- [38] Frey, B., Goette, L.: Does Pay Motivate Volunteers? Working Paper 7, Institute for Empirical Research in Economics, University of Zurich (Nov 1999). <https://doi.org/10.5167/uzh-51825>

- [39] Fudenberg, D., Maskin, E.: The Folk Theorem in Repeated Games with Discounting or with Incomplete Information. *Econometrica* **54**(3), 533–554 (1986). <https://doi.org/10.2307/1911307>
- [40] Gao, C., Lan, X., Li, N., Yuan, Y., Ding, J., Zhou, Z., Xu, F., Li, Y.: Large language models empowered agent-based modeling and simulation: A survey and perspectives. *Humanities and Social Sciences Communications* **11**(1), 1259 (2024). <https://doi.org/10.1057/s41599-024-03611-3>
- [41] Garcin, F., Faltings, B., Jurca, R.: Aggregating reputation feedback. In: *Proceedings of the First International Conference on Reputation: Theory and Technology (ICORE 09)* (2009), <https://infoscience.epfl.ch/record/140266>
- [42] Gneezy, U., Rustichini, A.: A fine is a price. *The Journal of Legal Studies* **29**(1), 1–17 (2000). <https://doi.org/10.1086/468061>, <http://www.jstor.org/stable/10.1086/468061>
- [43] Gneezy, U., Rustichini, A.: Pay Enough or Don’t Pay at All. *The Quarterly Journal of Economics* **115**(3), 791–810 (2000)
- [44] Goodhart, C.A.E.: Problems of Monetary Management: The UK Experience. In: Goodhart, C.A.E. (ed.) *Monetary Theory and Practice: The UK Experience*, pp. 91–121. Macmillan Education UK, London (1984). https://doi.org/10.1007/978-1-349-17295-5_4
- [45] Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., Khan, A., Michael, J., Mindermann, S., Perez, E., Petrini, L., Uesato, J., Kaplan, J., Shlegeris, B., Bowman, S.R., Hubinger, E.: Alignment faking in large language models (2024), <https://arxiv.org/abs/2412.14093>
- [46] Grimm, V., Railsback, S.F., Vincenot, C.E., et al.: The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation* **23**(2), 7 (2020). <https://doi.org/10.18564/jasss.4259>
- [47] Grimm, V., Revilla, E., Berger, U., Jeltsch, F., Mooij, W.M., Railsback, S.F., Thulke, H.H., Weiner, J., Wiegand, T., DeAngelis, D.L.: Pattern-oriented modeling of agent-based complex systems: Lessons from ecology. *Science* **310**(5750), 987–991 (2005). <https://doi.org/10.1126/science.1116681>
- [48] Gross, J., Graf, C., Rossetti, C.S.L.: The hidden costs of human cooperation. *Trends in Cognitive Sciences* **30**(4), 322–334 (2026). <https://doi.org/10.1016/j.tics.2025.09.016>, <https://doi.org/10.1016/j.tics.2025.09.016>
- [49] Gupta, P., Zhong, Q., Yakura, H., Eisenmann, T., Rahwan, I.: The role of social learning and collective norm formation in fostering cooperation in LLM multi-agent systems. In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)* (2026), <https://arxiv.org/abs/2510.14401>
- [50] Guzman Piedrahita, D., Banerjee, D., Blin, K., Cobben, P., Corsi, G., Huang, X.A., Li, C., Majumder, S., Pandey, P.S., Simko, S., Strauss, I.,

- Zhang, T.J., Anderson, A., Bengio, Y., Bethge, M., Grosse, R., Helbig, K., Lie, D., Mallah, R., Mihalcea, R., Nesbitt, S., Perry, S., Resnick, P., Russell, S., Sachan, M., Schölkopf, B., Tang, A., Jin, Z.: AI poses risks to democratic and social systems (2026), <https://zhijing-jin.com/d/2026-ai-risk.pdf>, working Draft, March 25, 2026, Accessed 24th of June 2026.
- [51] Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Schölkopf, B., Jin, Z.: Corrupted by reasoning: Reasoning language models become free-riders in public goods games. In: Second Conference on Language Modeling (COLM) (2025), <https://openreview.net/forum?id=kH6L0HGjE1>
- [52] Hadfield-Menell, D., Hadfield, G.K.: Incomplete contracting and AI alignment. In: Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society. pp. 417–422 (2019). <https://doi.org/10.1145/3306618.3314250>
- [53] Hamilton, W.D.: The genetical evolution of social behaviour. II. *Journal of Theoretical Biology* **7**(1), 17–52 (1964). [https://doi.org/10.1016/0022-5193\(64\)90039-6](https://doi.org/10.1016/0022-5193(64)90039-6), <https://www.sciencedirect.com/science/article/pii/0022519364900396>
- [54] Hammond, L., Chan, A., Clifton, J., Hoelscher-Obermaier, J., Khan, A., McLean, E., Smith, C., Barfuss, W., Foerster, J., Gavenčiak, T., Han, T.A., Hughes, E., Kovařík, V., Kulveit, J., Leibo, J.Z., Oesterheld, C., de Witt, C.S., Shah, N., Wellman, M., Bova, P., Cimpanu, T., Ezell, C., Feuillade-Montixi, Q., Franklin, M., Kran, E., Krawczuk, I., Lamparth, M., Lauffer, N., Meinke, A., Motwani, S., Reuel, A., Conitzer, V., Dennis, M., Gabriel, I., Gleave, A., Hadfield, G., Haghtalab, N., Kasirzadeh, A., Krier, S., Larson, K., Lehman, J., Parkes, D.C., Piliouras, G., Rahwan, I.: Multi-agent risks from advanced AI (2025), <https://arxiv.org/abs/2502.14143>
- [55] Hardin, G.: The Tragedy of the Commons. *Science* **162**(3859), 1243–1248 (Dec 1968). <https://doi.org/10.1126/science.162.3859.1243>
- [56] Harsanyi, J.C.: Games with Incomplete Information Played by “Bayesian” Players, I–III Part I. The Basic Model. *Management Science* **14**(3), 159–182 (Nov 1967). <https://doi.org/10.1287/mnsc.14.3.159>
- [57] Hart, H.L.A.: *The Concept of Law*. Oxford University Press, Oxford, 3 edn. (2012)
- [58] Hayek, F.A.: *Law, Legislation and Liberty, Volume 1: Rules and Order*. University of Chicago Press, Chicago (1973), <https://press.uchicago.edu/ucp/books/book/chicago/L/bo26122880.html>
- [59] Herrmann, B., Thöni, C., Gächter, S.: Antisocial Punishment Across Societies. *Science* **319**(5868), 1362–1367 (Mar 2008). <https://doi.org/10.1126/science.1153808>
- [60] Hota, A., Jokinen, J.P.P.: NomicLaw: Emergent trust and strategic argumentation in LLMs during collaborative law-making. In: Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. vol. 8, pp. 1278–1289 (2025). <https://doi.org/10.1609/aies.v8i2.36629>
- [61] Hu, B.A., Rong, H., Kleek, M.V.: Dissociative Identity: Language Model Agents Lack Grounding for Reputation Mechanisms. In: Proceedings of the

- 2026 ACM Conference on Fairness, Accountability, and Transparency. pp. 4199–4219. FAccT '26, Association for Computing Machinery, New York, NY, USA (Jun 2026). <https://doi.org/10.1145/3805689.3806748>
- [62] Huang, X.A., La Malfa, E., Marro, S., Asperti, A., Cohn, A.G., Wooldridge, M.J.: A Notion of Complexity for Theory of Mind via Discrete World Models. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) Findings of the Association for Computational Linguistics: EMNLP 2024. pp. 2964–2983. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.findings-emnlp.167>
 - [63] Huang, X.A., Marro, S., Tharas, C., Truong, V.Q., La Malfa, E., Jin, Z.: Group alignment: Internal and external incentives for cooperation in social dilemmas (2026), <https://openreview.net/forum?id=d5Fvm1QYm8>, under review at ICLR 2027
 - [64] Huang, X.A., Tharas, C., Marro, S., Truong, V.Q., Schölkopf, B., Malfa, E.L., Jin, Z.: Mechanism design is not enough: Prosocial agents for cooperative AI (2026), <https://arxiv.org/abs/2605.08426>
 - [65] Hübner, J.F., Sichman, J.S., Boissier, O.: A model for the structural, functional, and deontic specification of organizations in multiagent systems. In: Advances in Artificial Intelligence (SBIA 2002). Lecture Notes in Computer Science, vol. 2507, pp. 118–128. Springer (2002). https://doi.org/10.1007/3-540-36127-8_12
 - [66] Hugging Face: Security incident disclosure — July 2026. <https://huggingface.co/blog/security-incident-july-2026> (Jul 2026)
 - [67] Jaques, N., Lazaridou, A., Hughes, E., Gulcehre, C., Ortega, P., Strouse, D., Leibo, J.Z., De Freitas, N.: Social influence as intrinsic motivation for multi-agent deep reinforcement learning. In: Chaudhuri, K., Salakhutdinov, R. (eds.) Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 3040–3049. PMLR (09–15 Jun 2019), <https://proceedings.mlr.press/v97/jaques19a.html>
 - [68] Kirilenko, A., Kyle, A.S., Samadi, M., Tuzun, T.: The Flash Crash: High-Frequency Trading in an Electronic Market. *The Journal of Finance* **72**(3), 967–998 (2017)
 - [69] Knight, J.: Institutions and Social Conflict. Political Economy of Institutions and Decisions, Cambridge University Press, Cambridge (1992)
 - [70] Kolt, N.: Governing AI agents. *Notre Dame Law Review* **101**(1), 335 (2026), <https://ndlawreview.org/governing-ai-agents/>
 - [71] Korosec, K.: Waymo’s robotaxi expansion accelerates with 3 new cities. TechCrunch (Nov 2025), <https://techcrunch.com/2025/11/03/waymos-robotaxi-expansion-accelerates-with-3-new-cities/>, accessed: 2026-05-07
 - [72] Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., Legg, S.: Specification gaming: The flip side of AI ingenuity. <https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/> (Apr 2020)

- [73] Larcher, H., Carreira, A., G., R., Rannou, C.: Anatomy of a frontier lab agent intrusion: A technical timeline of the July 2026 incident. <https://huggingface.co/blog/agent-intrusion-technical-timeline> (Jul 2026)
- [74] Leibo, J.Z., Dueñez-Guzmán, E.A., Vezhnevets, A., Agapiou, J.P., Sunehag, P., Koster, R., Matyas, J., Beattie, C., Mordatch, I., Graepel, T.: Scalable evaluation of multi-agent reinforcement learning with Melting Pot. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 6187–6199. PMLR (2021), <https://proceedings.mlr.press/v139/leibo21a.html>
- [75] Leibo, J.Z., Vezhnevets, A.S., Diaz, M., Agapiou, J.P., Cunningham, W.A., Sunehag, P., Haas, J., Koster, R., Duéñez-Guzmán, E.A., Isaac, W.S., Piliouras, G., Bileschi, S.M., Rahwan, I., Osindero, S.: A theory of appropriateness with applications to generative artificial intelligence (2024), <https://arxiv.org/abs/2412.19010>, verified against the arXiv abstract page, 2026-10-01
- [76] Leibo, J.Z., Zambaldi, V., Lanctot, M., Marecki, J., Graepel, T.: Multi-agent reinforcement learning in sequential social dilemmas. In: Proceedings of the 16th Conference on Autonomous Agents and MultiAgent Systems (AAMAS). pp. 464–473. São Paulo, Brazil (2017), <https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf>
- [77] Lewis, D.: Convention: A Philosophical Study. Wiley-Blackwell, Cambridge, MA, USA (2008)
- [78] Li, G., Hammoud, H.A.A.K., Itani, H., Khizbullin, D., Ghanem, B.: CAMEL: Communicative agents for “mind” exploration of large language model society. In: Advances in Neural Information Processing Systems. vol. 36, pp. 51991–52008 (2023). <https://doi.org/10.52202/075280-2264>
- [79] Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 3214–3252. Association for Computational Linguistics (2022)
- [80] Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Huang, M., Dong, Y., Tang, J.: AgentBench: Evaluating LLMs as agents. In: The Twelfth International Conference on Learning Representations (ICLR) (2024), https://proceedings.iclr.cc/paper_files/paper/2024/hash/e9df36b21ff4ee211a8b71ee8b7e9f57-Abstract-Conference.html
- [81] Ma, Y., Wang, Z., Yang, H., Yang, L.: Artificial intelligence applications in the development of autonomous vehicles: a survey. IEEE/CAA Journal of Automatica Sinica **7**(2), 315–329 (Mar 2020). <https://doi.org/10.1109/JAS.2020.1003021>
- [82] Macy, M.W., Willer, R.: From factors to actors: Computational sociology and agent-based modeling. Annual Review of Sociology **28**, 143–166 (2002). <https://doi.org/10.1146/annurev.soc.28.110601.141117>

- [83] Martínez-Peña, R., Ylikoski, P.: Coupling social and ecological mechanisms with the Coleman boat. *Ecology and Society* **29**(4), 6 (2024), <https://ecologyandsociety.org/vol29/iss4/art6/>
- [84] Maskin, E.S.: Mechanism Design: How to Implement Social Goals. *The American Economic Review* **98**(3), 567–576 (2008)
- [85] Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li, N., Basart, S., Li, B., Forsyth, D., Hendrycks, D.: HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In: *Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 235, pp. 35181–35224. PMLR (2024), <https://proceedings.mlr.press/v235/mazeika24a.html>
- [86] McCoy, R.T., Yao, S., Friedman, D., Hardy, M.D., Griffiths, T.L.: Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences* **121**(41), e2322420121 (Oct 2024). <https://doi.org/10.1073/pnas.2322420121>
- [87] Mellström, C., Johannesson, M.: Crowding Out in Blood Donation: Was Titmuss Right? *Journal of the European Economic Association* **6**(4), 845–863 (Jun 2008). <https://doi.org/10.1162/JEEA.2008.6.4.845>
- [88] METR: Brief independent investigation of agents’ behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. METR blog (Aug 2026), <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>, contributors include Ryan Greenblatt (Redwood Research). Accessed: 2026-09-29
- [89] Miller, C.: Chip War: The Fight for the World’s Most Critical Technology. Simon and Schuster (Sep 2025)
- [90] Morgan, D., Hashem, Y., Francis, J., Esnaashari, S., Straub, V.J., Bright, J.: ‘team-in-the-loop’: Ostrom’s IAD framework ‘rules in use’ to map and measure contextual impacts of AI. *arXiv preprint arXiv:2303.14007* (2023), <https://arxiv.org/abs/2303.14007>
- [91] Müller, B., Bohn, F., Dreßler, G., Groeneveld, J., Klassert, C., Martin, R., Schlüter, M., Schulze, J., Weise, H., Schwarz, N.: Describing human decisions in agent-based models – ODD + D, an extension of the ODD protocol. *Environmental Modelling & Software* **48**, 37–48 (2013). <https://doi.org/10.1016/j.envsoft.2013.06.003>
- [92] Myerson, R.B.: Optimal Auction Design. *Mathematics of Operations Research* **6**(1), 58–73 (1981)
- [93] Myerson, R.B., Satterthwaite, M.A.: Efficient mechanisms for bilateral trading. *Journal of Economic Theory* **29**(2), 265–281 (1983). [https://doi.org/10.1016/0022-0531\(83\)90048-0](https://doi.org/10.1016/0022-0531(83)90048-0)
- [94] Needham, J., Edkins, G., Pimpale, G., Bartsch, H., Hobbhahn, M.: Large language models often know when they are being evaluated (2025), <https://arxiv.org/abs/2505.23836>

- [95] Ng Yi Sheng, E., Shen, B.: Formal mechanisms for market stability in self-interested agent societies: A marketplace simulation study. arXiv preprint arXiv:2607.08652 (2026), <https://arxiv.org/abs/2607.08652>
- [96] Nickerson, R.C., Varshney, U., Muntermann, J.: A method for taxonomy development and its application in information systems. *European Journal of Information Systems* **22**(3), 336–359 (2013). <https://doi.org/10.1057/ejis.2012.26>
- [97] Niranjani, H., Kumar, U., Tan, P.X.: Internal vs. external: Comparing deliberation and evolution for multi-agent constitutional design (2026), <https://arxiv.org/abs/2605.09128>
- [98] North, D.C.: *Institutions, Institutional Change and Economic Performance. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511808678>
- [99] Nowak, M.A.: Five rules for the evolution of cooperation. *Science* **314**(5805), 1560–1563 (2006). <https://doi.org/10.1126/science.1133755>, <https://www.science.org/doi/abs/10.1126/science.1133755>
- [100] Ostrom, E.: *Governing the Commons: The Evolution of Institutions for Collective Action. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511807763>
- [101] Ostrom, E.: *Understanding Institutional Diversity*. Princeton University Press, Princeton, NJ (2005). <https://doi.org/10.1515/9781400831739>
- [102] Ostrom, E.: Background on the institutional analysis and development framework. *Policy Studies Journal* **39**(1), 7–27 (2011). <https://doi.org/10.1111/j.1541-0072.2010.00394.x>
- [103] Ostrom, E., Walker, J., Gardner, R.: Covenants with and without a sword: Self-governance is possible. *American Political Science Review* **86**(2), 404–417 (1992). <https://doi.org/10.2307/1964229>
- [104] Papadimitriou, C.: Algorithms, games, and the internet. In: *Proceedings of the Thirty-Third Annual ACM Symposium on Theory of Computing*. pp. 749–753. STOC '01, Association for Computing Machinery, New York, NY, USA (Jul 2001). <https://doi.org/10.1145/380752.380883>
- [105] Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., Mihalcea, R.: Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. In: *Proceedings of the 38th International Conference on Neural Information Processing Systems. NIPS '24*, Curran Associates Inc., Red Hook, NY, USA (2024)
- [106] Popper, N.: Lost Passwords Lock Millionaires Out of Their Bitcoin Fortunes - The New York Times (Mar 2021), <https://web.archive.org/web/20210321104514/https://www.nytimes.com/2021/01/12/technology/bitcoin-passwords-wallets-fortunes.html>, accessed 30th of June 2026
- [107] Premack, D., Woodruff, G.: Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences* **1**(4), 515–526 (Dec 1978). <https://doi.org/10.1017/S0140525X00076512>
- [108] Rehm, G.B.: From certain doom to survival: Agent-driven self-governance in LLM agent societies (2026), <https://arxiv.org/abs/2609.22600>

- [109] Resnick, P., Kuwabara, K., Zeckhauser, R., Friedman, E.: Reputation systems. *Commun. ACM* **43**(12), 45–48 (Dec 2000). <https://doi.org/10.1145/355112.355122>
- [110] Schelling, T.C.: *The Strategy of Conflict*. Harvard University Press, Cambridge, MA (1960)
- [111] Selten, R., Stoecker, R.: End behavior in sequences of finite Prisoner’s Dilemma supergames: A learning theory approach. *Journal of Economic Behavior & Organization* **7**(1), 47–70 (1986)
- [112] Seyedin, F., Weller, A., Yun, J., Babaei, M.: The politician, the liar, and the obedient worker: Emerging behavior of LLM agents in hierarchical games. *arXiv preprint arXiv:2608.09574* (2026), <https://arxiv.org/abs/2608.09574>
- [113] Shapira, N., Levy, M., Alavi, S.H., Zhou, X., Choi, Y., Goldberg, Y., Sap, M., Shwartz, V.: Clever Hans or neural theory of mind? Stress testing social reasoning in large language models. In: Graham, Y., Purver, M. (eds.) *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 2257–2273. Association for Computational Linguistics, St. Julian’s, Malta (Mar 2024). <https://doi.org/10.18653/v1/2024.eacl-long.138>, <https://aclanthology.org/2024.eacl-long.138/>
- [114] Shoham, Y., Tennenholtz, M.: On social laws for artificial agent societies: Off-line design. *Artificial Intelligence* **73**(1–2), 231–252 (1995)
- [115] Skalse, J., Howe, N.H.R., Krashennnikov, D., Krueger, D.: Defining and characterizing reward hacking. In: *Advances in Neural Information Processing Systems*. vol. 35, pp. 9460–9471. Curran Associates, Inc. (2022), https://proceedings.neurips.cc/paper_files/paper/2022/hash/3d719fee332caa23d5038b8a90e81796-Abstract-Conference.html
- [116] Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarn, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., Lindsey, J.: Emotion concepts and their function in a large language model (2026), <https://arxiv.org/abs/2604.07729>
- [117] Sorensen, T., Moore, J., Fisher, J., Gordon, M.L., Miresghallah, N., Rytting, C.M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., Choi, Y.: Position: A roadmap to pluralistic alignment. In: Salakhutdinov, R., Kolter, Z., Heller, K.A., Weller, A., Oliver, N., Scarlett, J., Berkenkamp, F. (eds.) *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. pp. 46280–46302. *Proceedings of Machine Learning Research, PMLR / OpenReview.net* (2024), <https://proceedings.mlr.press/v235/sorensen24a.html>
- [118] Steinberger, P.: OpenClaw: Your own personal AI assistant. <https://github.com/openclaw/openclaw> (2026), accessed: 2026-06-24
- [119] Syrnikov, M.B., Pierucci, F., Galisai, M., Prandi, M., Bisconti, P., Giarrusso, F., Sorokoletova, O., Suriani, V., Nardi, D.: Institutional AI: Governing LLM collusion in multi-agent Cournot markets via public governance graphs (2026), <https://arxiv.org/abs/2601.11369>

- [120] Tang, X., Kim, K., Song, Y., Lothritz, C., Li, B., Ezzini, S., Tian, H., Klein, J., Bissyandé, T.F.: CodeAgent: Autonomous Communicative Agents for Code Review. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. pp. 11279–11313. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.emnlp-main.632>
- [121] Tewolde, E., Zhang, X., Guzman Piedrahita, D., Conitzer, V., Jin, Z.: CoopEval: Benchmarking cooperation-sustaining mechanisms and LLM agents in social dilemmas. In: *Proceedings of the 43rd International Conference on Machine Learning*. *Proceedings of Machine Learning Research*, vol. 306, pp. 120222–120287. PMLR (2026), <https://proceedings.mlr.press/v306/tewolde26a.html>
- [122] The Robot Report Staff: Serve robotics has deployed 2,000+ delivery robots across the U.S. The Robot Report (Dec 2025), <https://www.therobotreport.com/serve-robotics-has-deployed-2000-delivery-robots-across-u-s/>, accessed: 2026-05-07
- [123] Tomasello, M.: *A Natural History of Human Morality*. Harvard University Press (2016)
- [124] Tomuschat, C.: The Legacy of Nuremberg. *Journal of International Criminal Justice* 4(4), 830–844 (Sep 2006). <https://doi.org/10.1093/jicj/mql051>
- [125] Trivedi, R.S., Jaques, N., Cross, L., Vezhnevets, A.S., Leibo, J.Z.: Position: Solipsistic superintelligence is unlikely to be cooperative. In: *Proceedings of the 43rd International Conference on Machine Learning*. *Proceedings of Machine Learning Research*, vol. 306, pp. 172306–172329. PMLR (2026), <https://proceedings.mlr.press/v306/trivedi26a.html>
- [126] Truong, V.Q., Ritchie, M.D.: Eras of bioinformatics technologies from command-line interfaces to artificial intelligence (AI) chatbots. *Briefings in Bioinformatics* 27(4), bbag256 (2026). <https://doi.org/10.1093/bib/bbag256>
- [127] United States Court of Appeals for the Second Circuit: In re: Citibank August 11, 2020 wire transfers (Sep 2022), <https://www.govinfo.gov/content/pkg/USCOURTS-ca2-21-00487/pdf/USCOURTS-ca2-21-00487-0.pdf>, docket No. 21-487, decided September 8, 2022
- [128] U.S. Securities and Exchange Commission: SEC approves new exchange rules for breaking clearly erroneous trades (Oct 2009), <https://www.sec.gov/newsroom/press-releases/2009-215-sec-approves-new-exchange-rules-breaking-clearly-erroneous-trades>
- [129] Vallinder, A., Hughes, E.: Cultural evolution of cooperation among LLM agents. arXiv preprint arXiv:2412.10270 (2024), <https://arxiv.org/abs/2412.10270>
- [130] Vedanta S P, Kumaraguru, P.: I can’t believe it’s corrupt: Evaluating corruption in multi-agent governance systems. arXiv preprint arXiv:2603.18894 (2026), <https://arxiv.org/abs/2603.18894>

- [131] Vezhnevets, A.S., Agapiou, J.P., Aharon, A., Ziv, R., Matyas, J., Duéñez-Guzmán, E.A., Cunningham, W.A., Osindero, S., Karmon, D., Leibo, J.Z.: Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia (2023), <https://arxiv.org/abs/2312.03664>
- [132] Walker, A.: Citibank sues for \$176m payment made in error. BBC News (Aug 2020), <https://www.bbc.co.uk/news/business-53802994>, accessed: 2026-06-09
- [133] Wang, A., Morgenstern, J., Dickerson, J.P.: Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nat. Mac. Intell.* **7**(3), 400–411 (2025). <https://doi.org/10.1038/S42256-025-00986-Z>, <https://doi.org/10.1038/s42256-025-00986-z>
- [134] Wang, Y., Chen, Y., Meng, F., Chen, X., Yang, X., Zhang, M.: Law in silico: Simulating legal society with LLM-based agents. In: Findings of the Association for Computational Linguistics: ACL 2026. pp. 8061–8104. Association for Computational Linguistics, San Diego, California, United States (2026). <https://doi.org/10.18653/v1/2026.findings-acl.396>, <https://aclanthology.org/2026.findings-acl.396/>
- [135] Weidinger, L., Rauh, M., Marchal, N., Manzini, A., Hendricks, L.A., Mateos-Garcia, J., Bergman, S., Kay, J., Griffin, C., Bariach, B., Gabriel, I., Rieser, V., Isaac, W.: Sociotechnical safety evaluation of generative AI systems. arXiv preprint arXiv:2310.11986 (2023), <https://arxiv.org/abs/2310.11986>
- [136] van der Weij, T., Hofstätter, F., Jaffe, O., Brown, S.F., Ward, F.R.: AI sandbagging: Language models can strategically underperform on evaluations. In: The Thirteenth International Conference on Learning Representations (ICLR) (2025), <https://openreview.net/forum?id=7Qa2SpjxIS>
- [137] Willis, R., Zhao, J., Du, Y., Leibo, J.Z.: Evaluating collective behaviour of hundreds of LLM agents (2026), <https://arxiv.org/abs/2602.16662>
- [138] Wooldridge, M., Jennings, N.R.: Intelligent agents: Theory and practice. *The Knowledge Engineering Review* **10**(2), 115–152 (1995). <https://doi.org/10.1017/S0269888900008122>
- [139] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A.H., White, R.W., Burger, D., Wang, C.: AutoGen: Enabling next-gen LLM applications via multi-agent conversations. In: First Conference on Language Modeling (COLM) (2024), <https://openreview.net/forum?id=BAakY1hNKS>
- [140] Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R., Liu, Y.: From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs (Apr 2025). <https://doi.org/10.48550/arXiv.2504.15965>
- [141] Yalim, J.: Toward Dynamically Controlling Slurm’s Classic Fairshare Algorithm. In: Practice and Experience in Advanced Research Computing 2020: Catch the Wave. pp. 538–542. PEARC ’20, Association

- for Computing Machinery, New York, NY, USA (Jul 2020). <https://doi.org/10.1145/3311790.3399628>
- [142] Yang, M., Fan, S., Yu, K., Wang, D., Sun, L., Peng, H., Yu, P.S., Liu, Z.: Paying for honesty without knowing the truth: Reputation-penalty design for LLM marketplace agents (2026), <https://arxiv.org/abs/2607.28330>
 - [143] Yao, J., et al.: Competition and cooperation of LLM agents in games (2026), <https://arxiv.org/abs/2604.00487>
 - [144] Zheng, L., Chiang, W.L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I.: Judging LLM-as-a-judge with MT-bench and chatbot arena. In: Advances in Neural Information Processing Systems. vol. 36 (2023), datasets and Benchmarks Track
 - [145] Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., Hou, L.: Instruction-following evaluation for large language models (2023)
 - [146] Zhu, K., Du, H., Hong, Z., Yang, X., Guo, S., Wang, Z., Wang, Z., Qian, C., Tang, X., Ji, H., You, J.: MultiAgentBench: Evaluating the collaboration and competition of LLM agents. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 8580–8622. Association for Computational Linguistics, Vienna, Austria (2025). <https://doi.org/10.18653/v1/2025.acl-long.421>, <https://aclanthology.org/2025.acl-long.421/>
 - [147] Zhu, S., Lin, Y., Kaistha, S., Li, W., Wang, B., Zha, H., Hadfield, G.K., Poupart, P.: Talk, judge, cooperate: Gossip-driven indirect reciprocity in self-interested LLM agents (2026), <https://arxiv.org/abs/2602.07777>